

EDA

2024-09-21

EDA

Load the data

```
current_folder <- getwd()
clean_data <- read.csv(file.path(current_folder, "data/clean_data.csv"))

head(clean_data)
```

##	AdjustedYear	ID	Sex	Team	Age	Weight	Height	NOC	Season	Sport
## 1	1896	135285	M	Greece	NA	NA	NA	GRE	Summer	Athletics
## 2	1896	39654	M	Greece	NA	NA	NA	GRE	Summer	Athletics
## 3	1896	35094	M	Greece	NA	NA	NA	GRE	Summer	Athletics
## 4	1896	129362	M	Hungary	23	NA	NA	HUN	Summer	Gymnastics
## 5	1896	56757	M	Hungary	NA	NA	NA	HUN	Summer	Gymnastics
## 6	1896	93996	M	Greece	NA	NA	NA	GRE	Summer	Shooting
##					Event	Medal	AthleteCountry			
## 1					Athletics Men's Triple Jump	<NA>	Greece			
## 2					Athletics Men's Marathon	<NA>	Greece			
## 3					Athletics Men's 1,500 metres	<NA>	Greece			
## 4					Gymnastics Men's Horizontal Bar	<NA>	Hungary			
## 5					Gymnastics Men's Parallel Bars	<NA>	Hungary			
## 6					Shooting Men's Military Rifle, 200 metres	<NA>	Greece			
##	Weight	Height	Ration	Country	Year	Home	Away			
## 1			NA	Greece	1896	Home				
## 2			NA	Greece	1896	Home				
## 3			NA	Greece	1896	Home				
## 4			NA	Greece	1896	Away				
## 5			NA	Greece	1896	Away				
## 6			NA	Greece	1896	Home				

EDA

We have data from 1896 to 2016:

```
# YEAR -----
c(min(clean_data$AdjustedYear), max(clean_data$AdjustedYear))
```

```
## [1] 1896 2016
```

The first summer Summer Olympics (1896) included 9 sports, the last (2016) ones include 34. The first summer Winter Olympics (1924) included 10 sports, the last (2014) ones include 15.

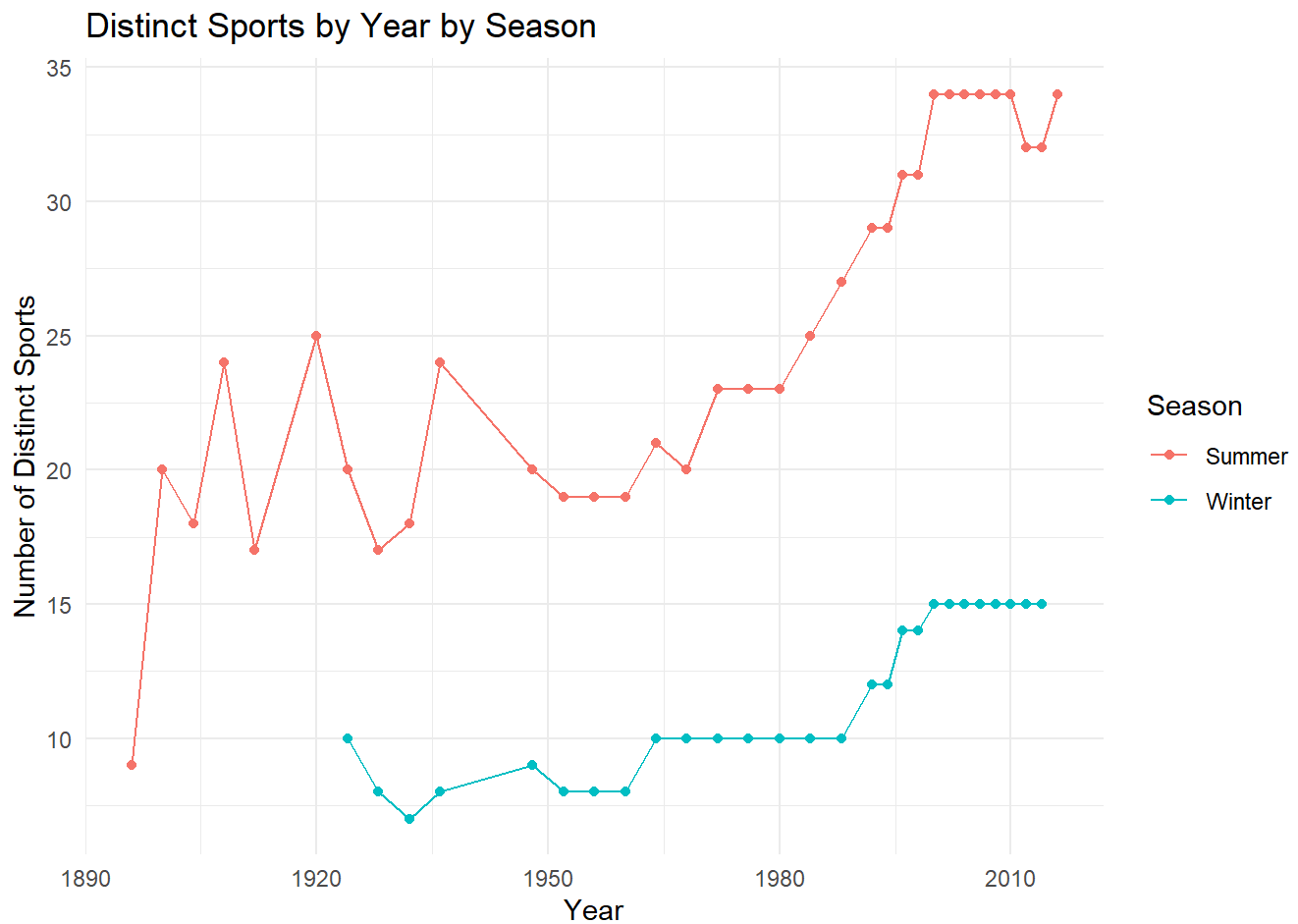
Sport

Q1: How many sports in each season? And how the #sports per season changed with time?

```
clean_data %>%
  group_by(Season, Year) %>% summarise(n_distinct(Sport), .groups='drop') %>%
  group_by(Season) %>%
  filter(Year==max(Year) | Year==min(Year))
```

```
## # A tibble: 4 × 3
## # Groups:   Season [2]
##   Season Year `n_distinct(Sport)`
##   <chr> <int>           <int>
## 1 Summer 1896             9
## 2 Summer 2016            34
## 3 Winter 1924            10
## 4 Winter 2014            15
```

```
clean_data %>%
  group_by(Year, Season) %>%
  summarize(NSports = n_distinct(Sport), .groups='drop') %>%
  ggplot(aes(x = Year, y = NSports, color=Season)) +
  geom_line() +
  geom_point() +
  labs(title = "Distinct Sports by Year by Season",
       x = "Year",
       y = "Number of Distinct Sports") +
  theme_minimal()
```



Q2: Which sports have been in the Olympic games every single year?

```
clean_data %>%
  group_by(Sport) %>%
  summarise(Years_Present = n_distinct(AdjustedYear)) %>%
  filter(Years_Present == n_distinct(clean_data$AdjustedYear)) %>%
  pull(Sport)
```

```
## [1] "Athletics" "Cycling" "Fencing" "Gymnastics" "Swimming"
```

Q3: Which sports have been played just once in the Olympics?

```
clean_data %>%
  group_by(Sport) %>%
  summarize(YearCount = n_distinct(Year),
            YearPlayed = first(Year[which(n_distinct(Year) == 1)])) %>%
  filter(YearCount == 1)
```

```
## # A tibble: 10 × 3
##   Sport          YearCount YearPlayed
##   <chr>          <int>     <int>
## 1 Aeronautics      1      1936
## 2 Basque Pelota    1      1900
## 3 Cricket          1      1900
## 4 Croquet          1      1900
## 5 Jeu De Paume     1      1908
## 6 Military Ski Patrol 1      1924
## 7 Motorboating     1      1908
## 8 Racquets         1      1908
## 9 Roque            1      1904
## 10 Rugby Sevens    1      2016
```

Height and Weight

Q1: Is there high correlation?

```
cor(clean_data$Weight, clean_data$Height, use = "complete.obs")
```

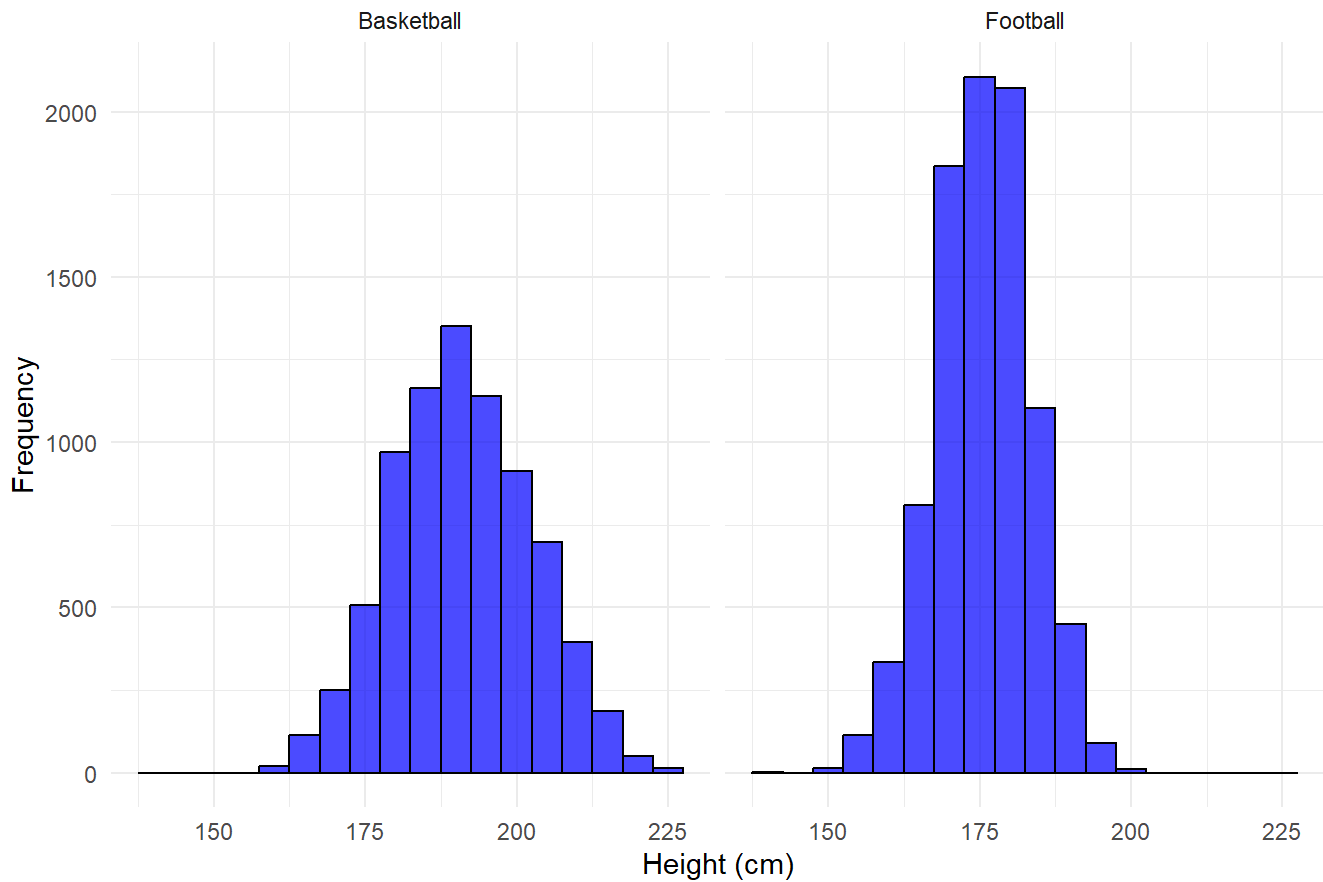
```
## [1] 0.7974434
```

Q2: Is the height distribution of athletes different in different sports?

```
clean_data %>%
  filter(Sport %in% c('Basketball', 'Football')) %>%
  ggplot(aes(x = Height)) +
  geom_histogram(binwidth = 5, fill = 'blue', color = 'black', alpha = 0.7) +
  facet_wrap(~ Sport) +
  theme_minimal() +
  labs(title = 'Height Distribution by Sport', x = 'Height (cm)', y = 'Frequency')
```

```
## Warning: Removed 5295 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

Height Distribution by Sport



Q3: T test to check if Basketball players are taller than Football players

```
basketball_players <- clean_data %>% filter(Sport == "Basketball")
football_players <- clean_data %>% filter(Sport == "Football")
t.test(basketball_players$Height, football_players$Height, alternate='greater')
```

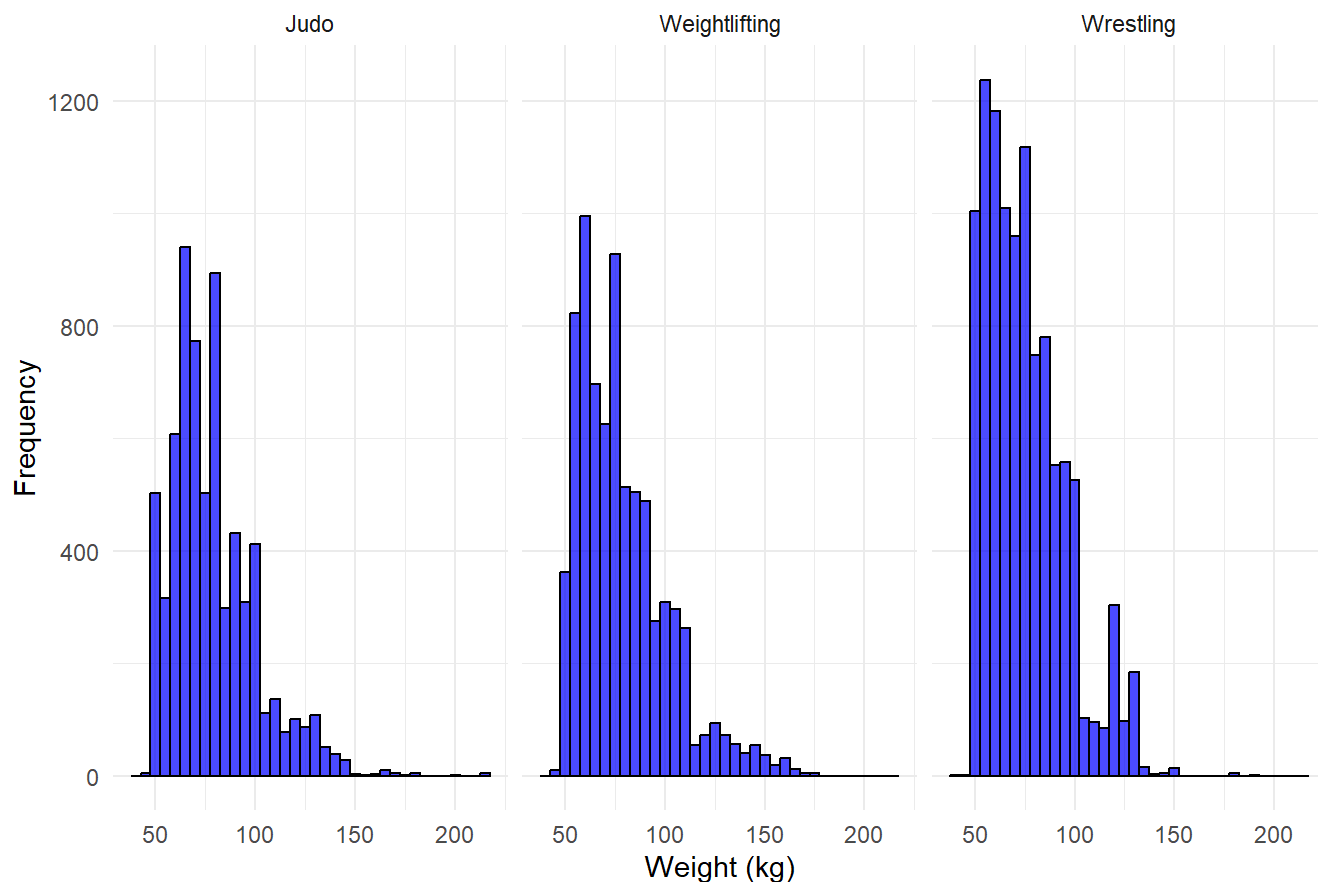
```
##
## Welch Two Sample t-test
##
## data: basketball_players$Height and football_players$Height
## t = 100.11, df = 13449, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 15.08184 15.68424
## sample estimates:
## mean of x mean of y
## 190.8951 175.5120
```

Q4: ANOVA Test: Is the weight distribution of athletes different in different sports?

```
clean_data %>%
  filter(Sport %in% c('Wrestling', 'Weightlifting', 'Judo')) %>%
  ggplot(aes(x = Weight)) +
  geom_histogram(binwidth = 5, fill = 'blue', color = 'black', alpha = 0.7) +
  facet_wrap(~ Sport) +
  theme_minimal() +
  labs(title = 'Weight Distribution by Sport', x = 'Weight (kg)', y = 'Frequency')
```

```
## Warning: Removed 4419 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

Weight Distribution by Sport



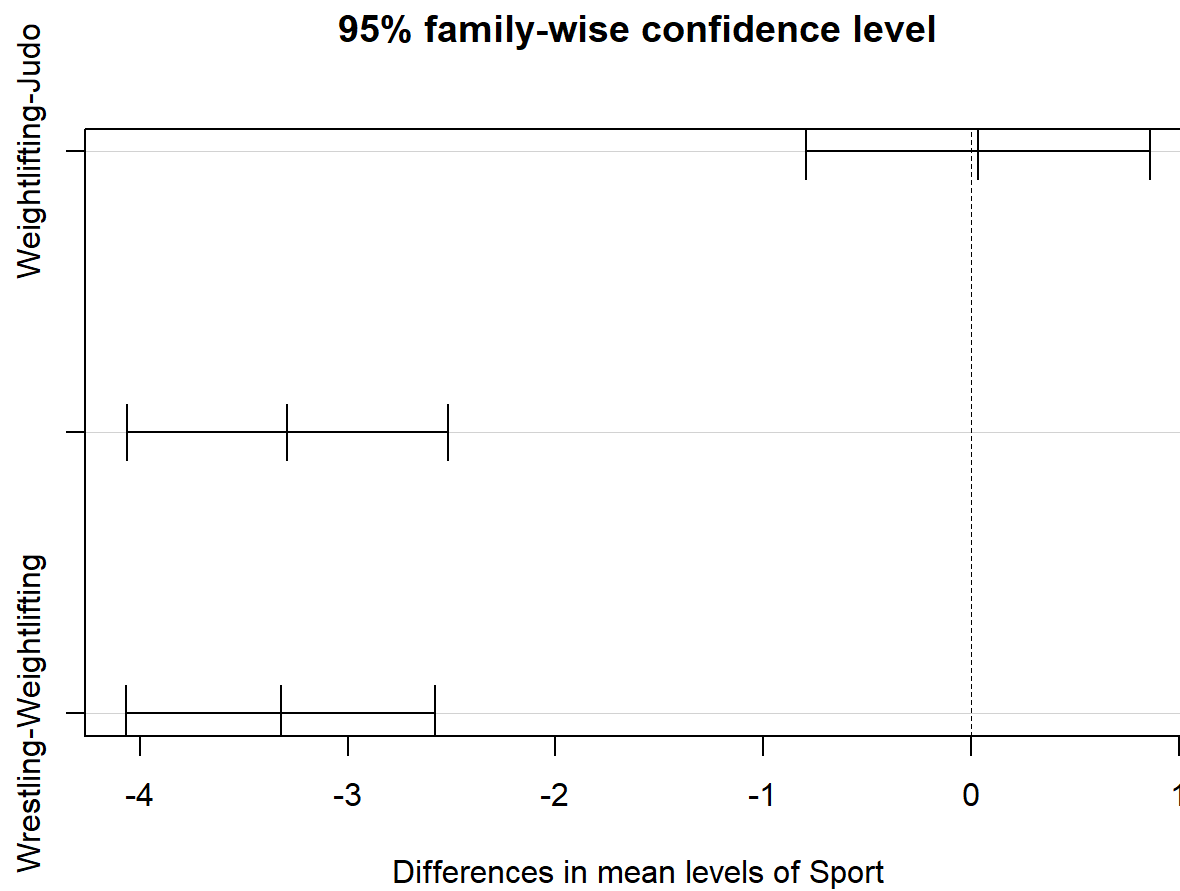
```
weight_diffs <- clean_data %>%
  filter(Sport %in% c('Wrestling', 'Weightlifting', 'Judo'))
anova <- aov(Weight~Sport, data=weight_diffs)
summary(anova)
```

```
##           Df    Sum Sq Mean Sq F value Pr(>F)
## Sport      2     66746   33373   74.59 <2e-16 ***
## Residuals 24981 11176370     447
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 4419 observations deleted due to missingness
```

```
TukeyHSD(anova, conf.level = 0.95)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Weight ~ Sport, data = weight_diffs)
##
## $Sport
##              diff          lwr          upr      p adj
## Weightlifting-Judo  0.03225926 -0.7956476  0.8601661 0.9954122
## Wrestling-Judo     -3.29045709 -4.0623631 -2.5185511 0.0000000
## Wrestling-Weightlifting -3.32271634 -4.0669577 -2.5784750 0.0000000
```

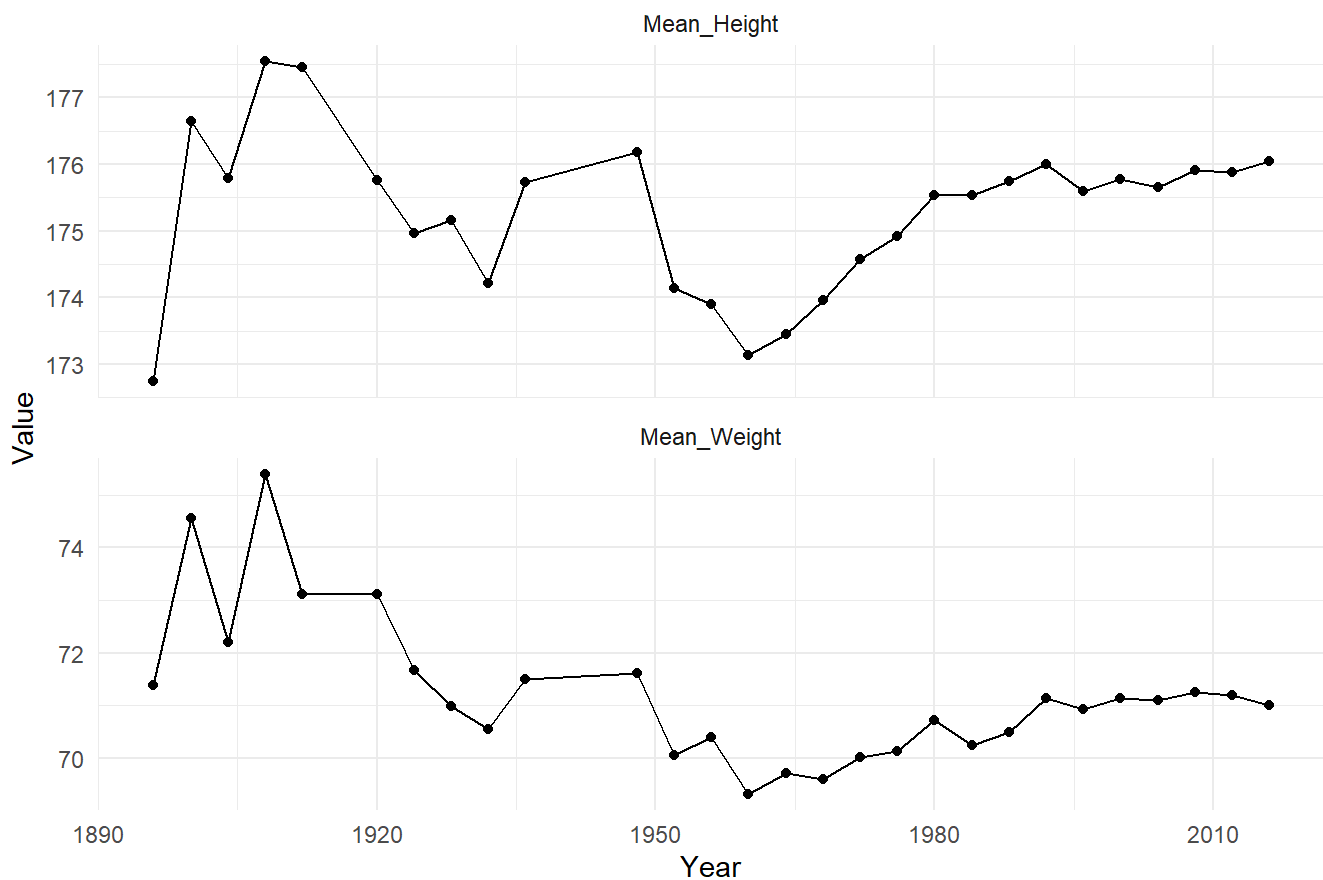
```
plot(TukeyHSD(anova, conf.level = 0.95))
```



Q5: Has weight and height changed over time? Which sports have seen largest differences in weight over time?

```
clean_data %>%
  group_by(AdjustedYear) %>%
  summarise(
    Mean_Height = mean(Height, na.rm=T),
    Mean_Weight = mean(Weight, na.rm=T)
  ) %>%
  pivot_longer(cols = c(Mean_Height, Mean_Weight), names_to = "Metric", values_to = "Value") %>%
  ggplot(aes(x = AdjustedYear, y = Value)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Metric, scales = "free_y", ncol = 1) +
  theme_minimal() +
  labs(title = 'Mean Height and Weight Over Years by Sport', x = 'Year', y = 'Value')
```

Mean Height and Weight Over Years by Sport



Q5: Which sports have seen largest differences in weight over time?

```
clean_data %>%
  group_by(Sport) %>%
  summarise(
    Mean_Height = mean(Height, na.rm=T),
    SD_Height = sd(Height, na.rm=T),
    Mean_Weight = mean(Weight, na.rm=T),
    SD_Weight = sd(Weight, na.rm=T)
  ) %>%
  arrange(desc(SD_Height))
```

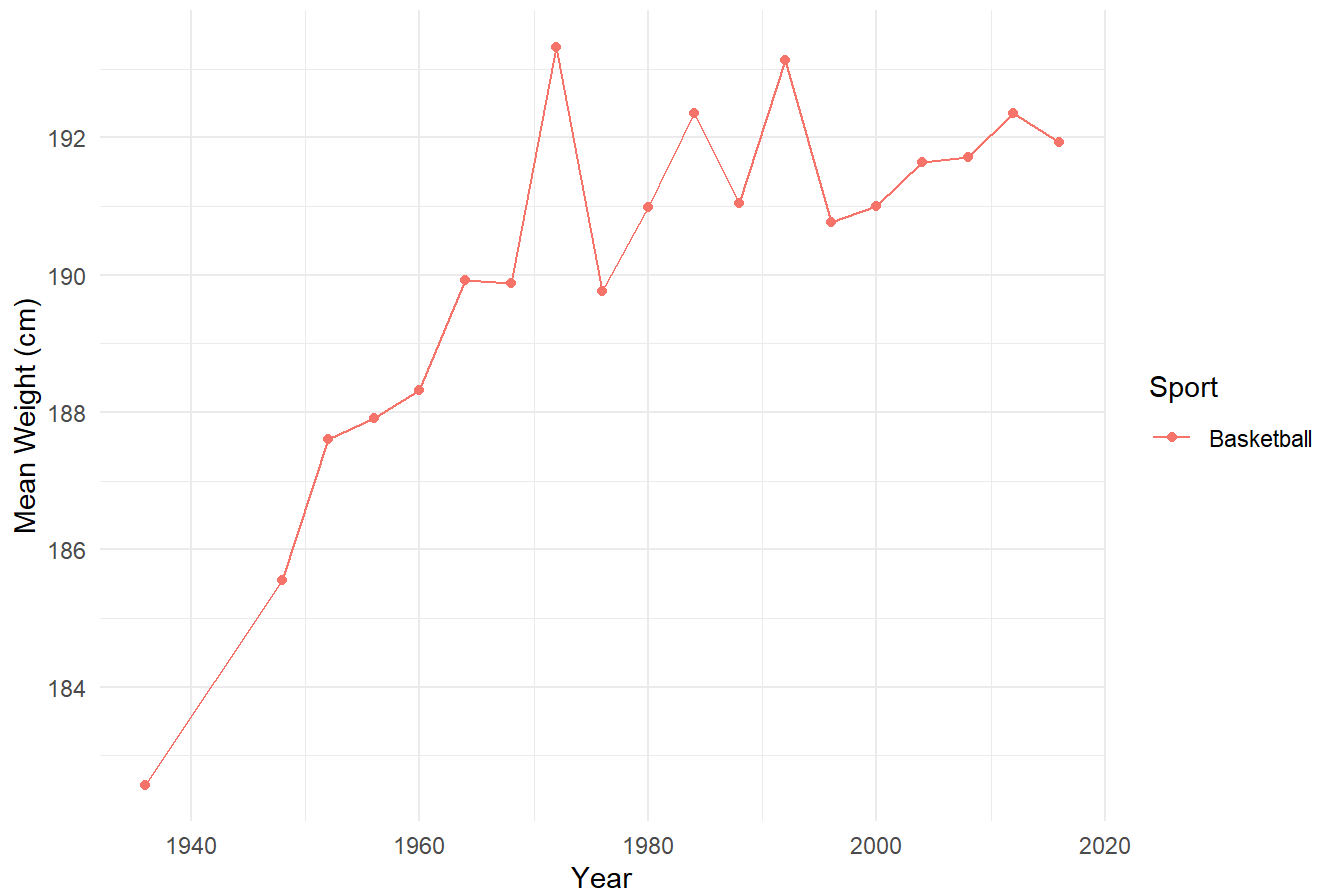


```
## # A tibble: 66 × 5
##   Sport      Mean_Height SD_Height Mean_Weight SD_Weight
##   <chr>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 Basketball    191.      11.4       85.8       14.8
## 2 Judo          174.      10.5       78.7       21.3
## 3 Volleyball    187.      10.4       78.9       11.7
## 4 Rugby Sevens  175.      10.2       78.9       14.6
## 5 Wrestling     172.      10.1       75.4       20.0
## 6 Figure Skating 168.      10.0       59.5       11.7
## 7 Taekwondo     176.       9.97       68.1       13.5
## 8 Weightlifting  168.       9.92       78.7       22.5
## 9 Rowing        184.       9.92       80.2       13.1
## 10 Swimming     179.       9.90       70.6       11.3
## # i 56 more rows
```

```
# Basketball has seen larger changes in the height of the athletes over time
# Plot
clean_data %>%
  filter(Sport %in% c('Basketball')) %>%
  group_by(AdjustedYear, Sport) %>%
  summarise( Mean_Height = mean(Height, na.rm=T) ) %>%
  ggplot(aes(x = AdjustedYear, y = Mean_Height, color=Sport)) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  labs(title = 'Mean Height Over Years by Sport', x = 'Year', y = 'Mean Weight (cm)')
```

```
## `summarise()` has grouped output by 'AdjustedYear'. You can override using the
## `.groups` argument.
```

Mean Height Over Years by Sport



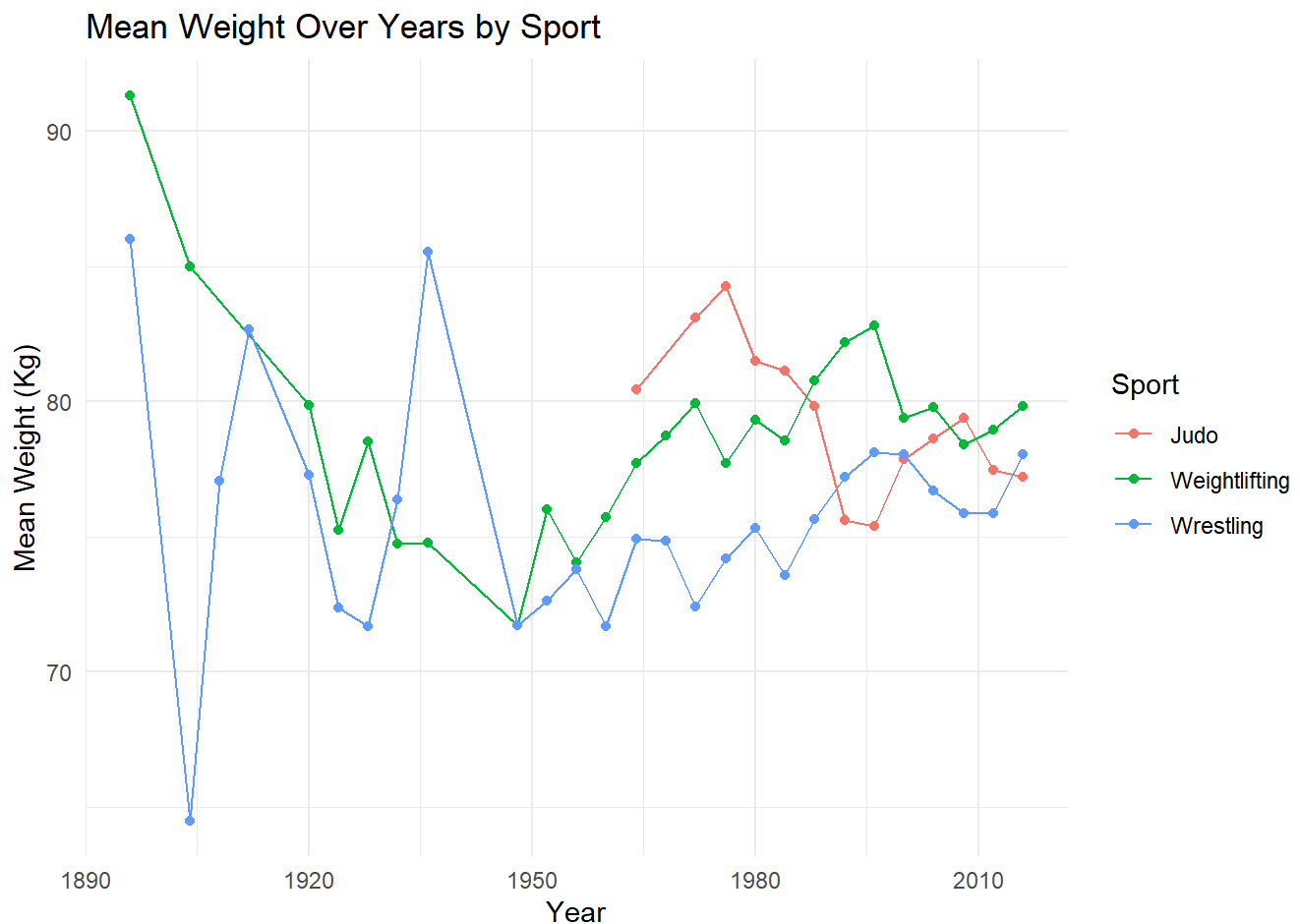
Q6: Which sports have seen largest differences in weight over time?

```
clean_data %>%
  group_by(Sport) %>%
  summarise(
    Mean_Weight = mean(Weight, na.rm=T),
    SD_Weight = sd(Weight, na.rm=T)
  ) %>%
  arrange(desc(SD_Weight))
```

```
## # A tibble: 66 x 3
##   Sport      Mean_Weight SD_Weight
##   <chr>          <dbl>     <dbl>
## 1 Weightlifting    78.7      22.5
## 2 Judo             78.7      21.3
## 3 Wrestling       75.4      20.0
## 4 Athletics       69.2      15.4
## 5 Basketball      85.8      14.8
## 6 Rugby Sevens    78.9      14.6
## 7 Lacrosse        76.7      13.8
## 8 Taekwondo       68.1      13.5
## 9 Shooting        74.0      13.2
## 10 Handball       81.4      13.1
## # i 56 more rows
```

```
# Weightlifting, Judo and Wrestling have seen larger changes in the weight of the athletes over time
# Plot
clean_data %>%
  filter(Sport %in% c('Weightlifting', 'Judo', 'Wrestling')) %>%
  group_by(AdjustedYear, Sport) %>%
  summarise( Mean_Weight = mean(Weight, na.rm=T) ) %>%
  ggplot(aes(x = AdjustedYear, y = Mean_Weight, color=Sport)) +
  geom_line() +
  geom_point() +
  theme_minimal() +
  labs(title = 'Mean Weight Over Years by Sport', x = 'Year', y = 'Mean Weight (Kg)')
```

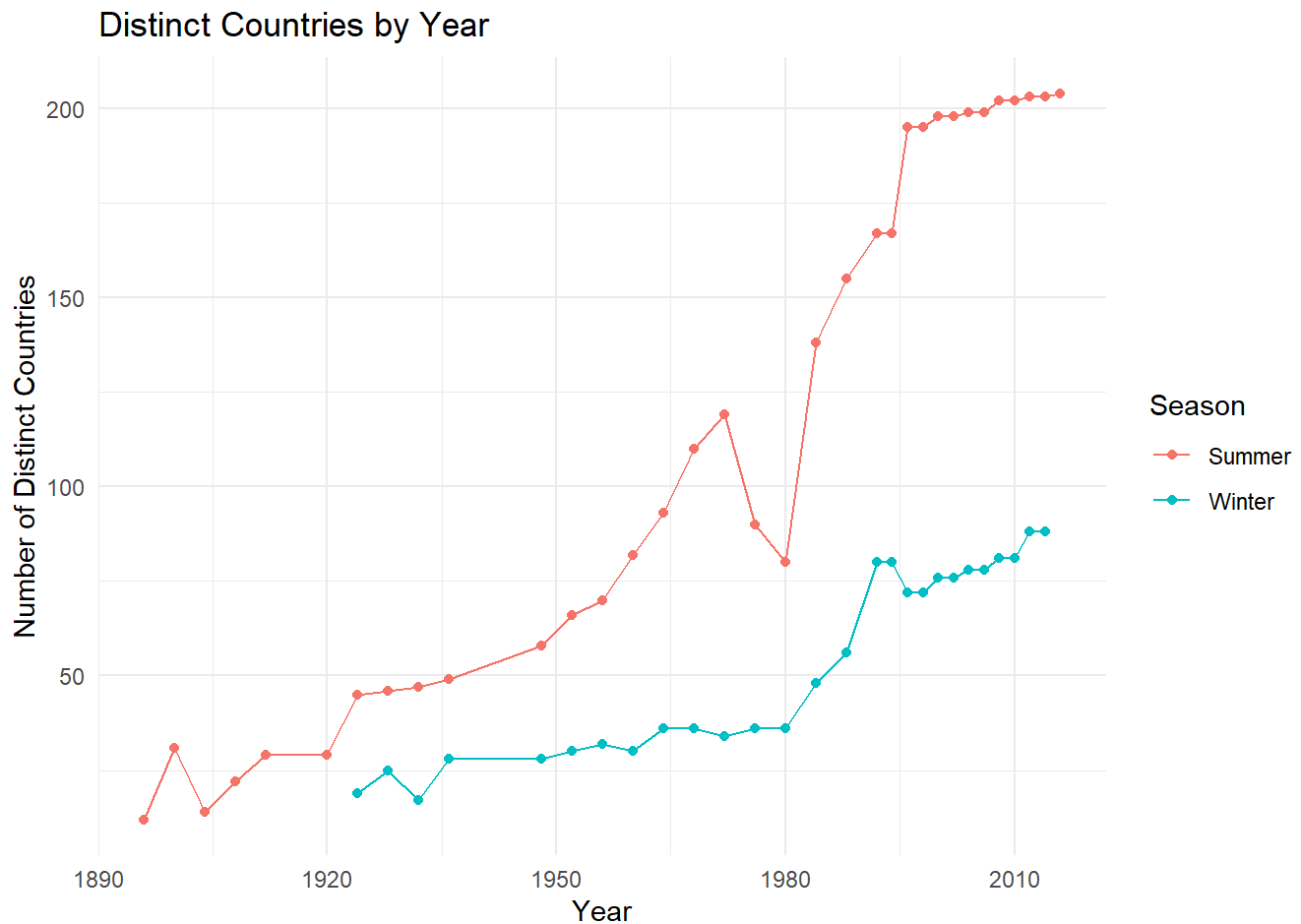
```
## `summarise()` has grouped output by 'AdjustedYear'. You can override using the
## `.groups` argument.
```



Country

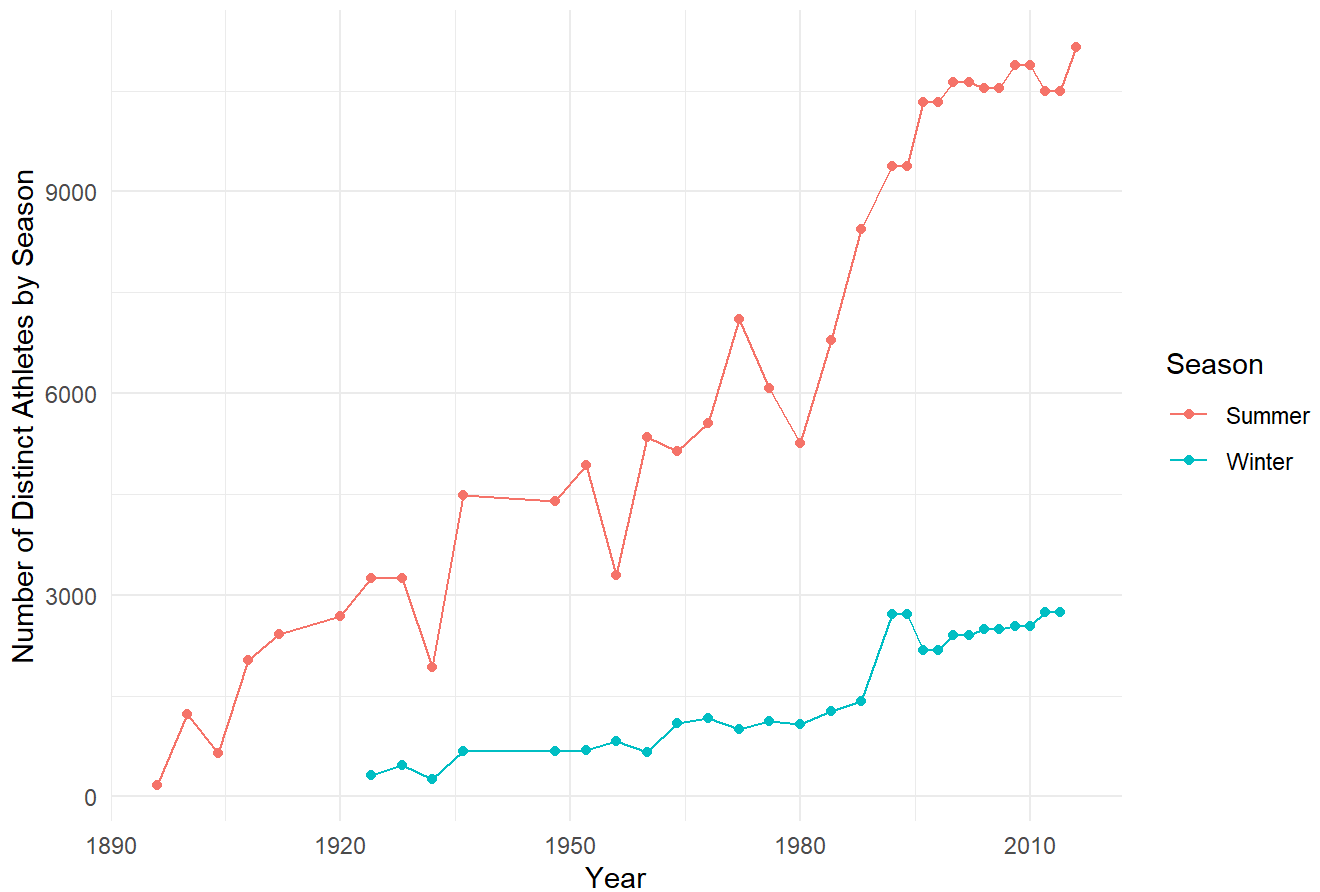
Q1: Has country participation increased with time?

```
clean_data %>%
  group_by(Year, Season) %>%
  summarize(NCountries = n_distinct(AthleteCountry), .groups='drop') %>%
  ggplot(aes(x = Year, y = NCountries, color=Season)) +
  geom_line() +
  geom_point() +
  labs(title = "Distinct Countries by Year",
       x = "Year",
       y = "Number of Distinct Countries") +
  theme_minimal()
```



```
clean_data %>%
  group_by(Year, Season) %>%
  summarize(NAthletes = n_distinct(ID), .groups='drop') %>%
  ggplot(aes(x = Year, y = NAthletes, color=Season)) +
  geom_line() +
  geom_point() +
  labs(title = "#Athletes by Year",
       x = "Year",
       y = "Number of Distinct Athletes by Season") +
  theme_minimal()
```

#Athletes by Year

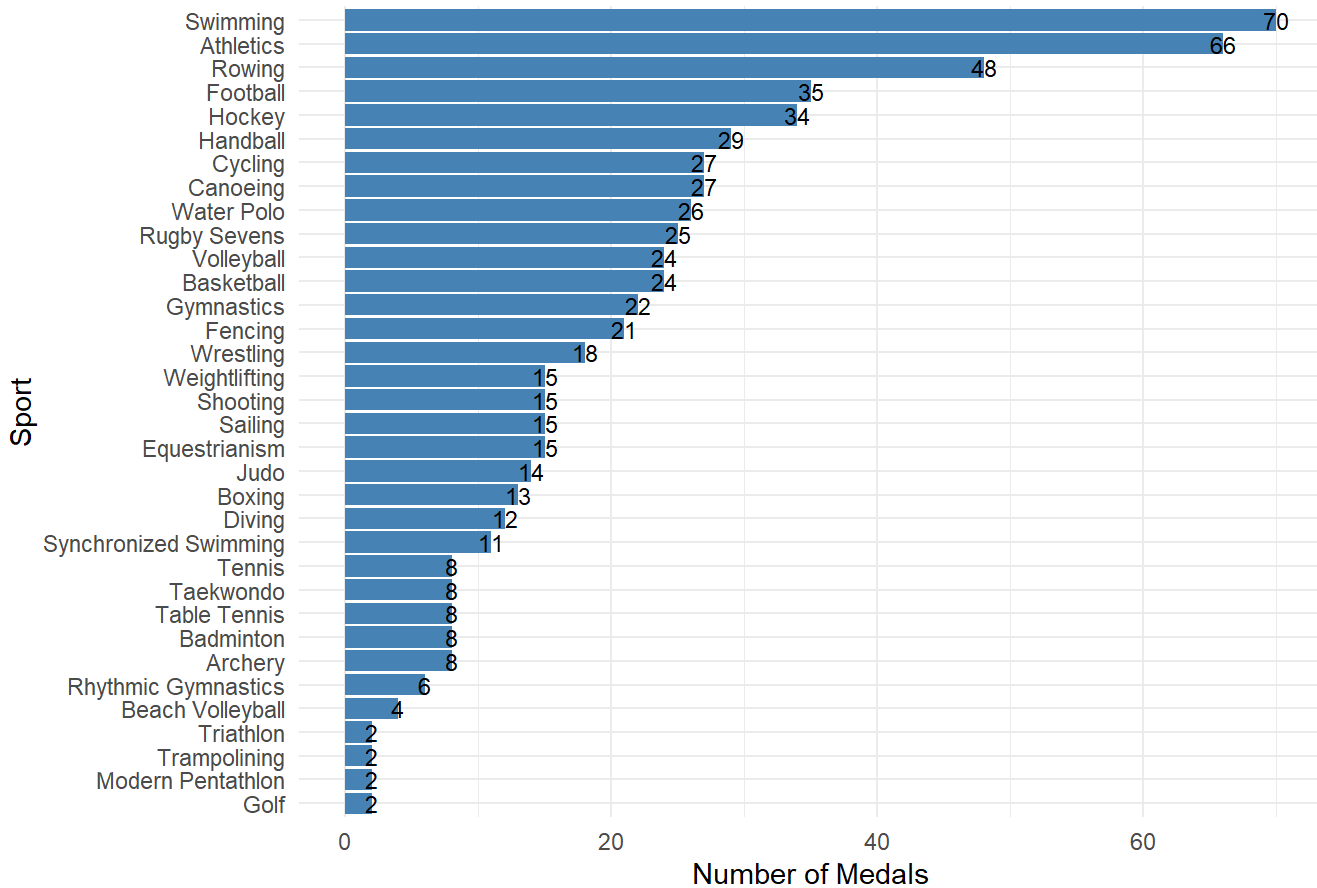


Country (and therefore #athletes) participation in the Olympic Games has increased significantly in the last 20 years. Higher participation in the Summer games compared to the Winter ones. A dip in the Summer Olympic Games in 1980 - Boycott of the Moscow Olympics by US to protest against Afghanistan invasion.

Q2: Which sports offer the more medals (gold medals) on average?

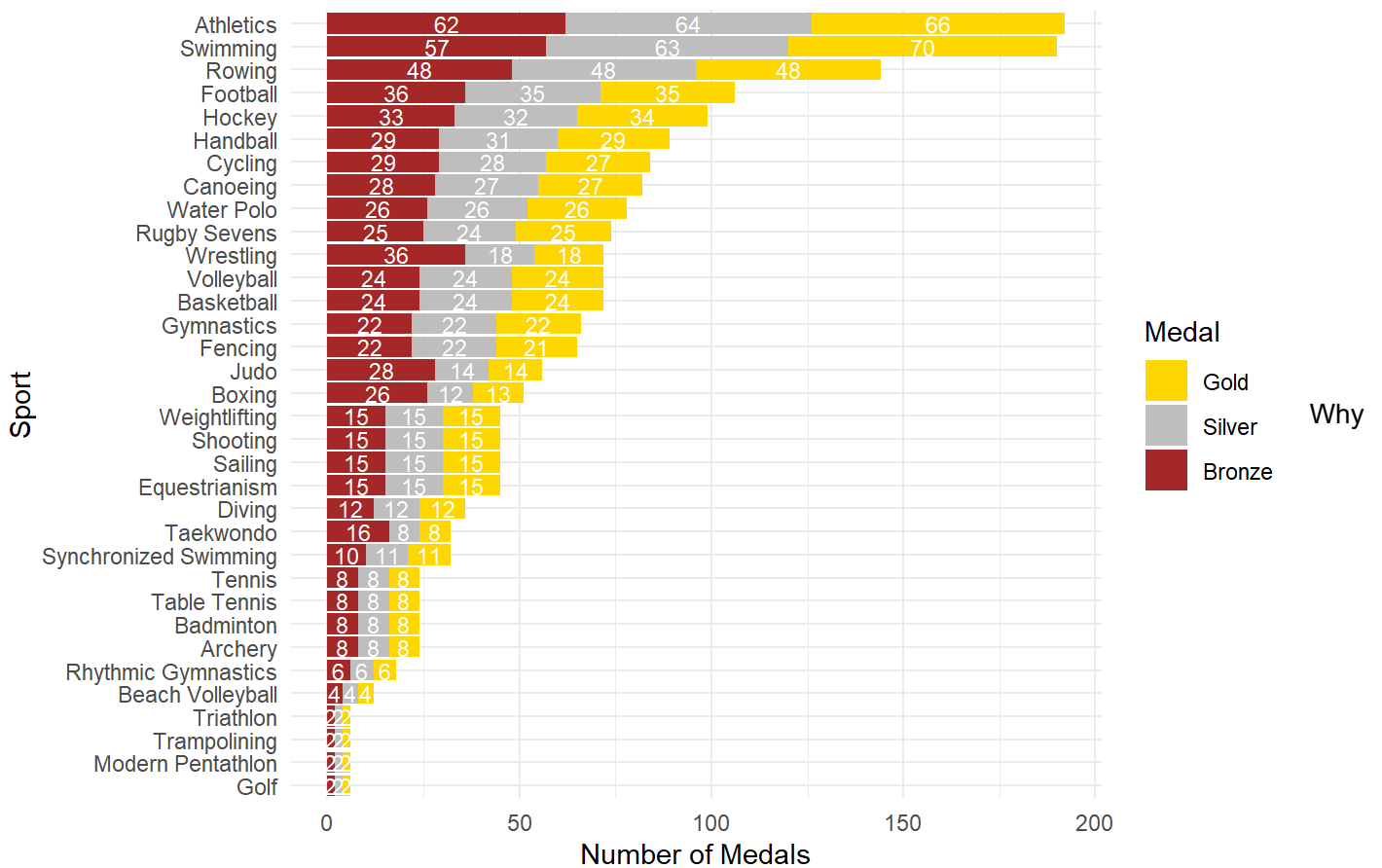
```
clean_data %>%
  filter(Year == 2016) %>%
  group_by(Sport) %>% summarize(NMedals = sum(!is.na(Medal) & Medal=='Gold')) %>% arrange(desc(NMedals)) %>%
  ggplot(aes(x = reorder(Sport, NMedals), y = NMedals)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = NMedals), size = 3, color = "black") +
  coord_flip() +
  labs(title = "Number of Medals by Sport",
       x = "Sport",
       y = "Number of Medals") +
  theme_minimal()
```

Number of Medals by Sport



```
clean_data %>%
  filter(Year == 2016 & !(is.na(Medal))) %>%
  group_by(Sport, Medal) %>%
  summarize(NMedals = sum(!is.na(Medal)), .groups = 'drop') %>%
  mutate(Medal = factor(Medal, levels = c("Gold", "Silver", "Bronze"))) %>%
  ggplot(aes(x = reorder(Sport, NMedals), y = NMedals, fill = Medal)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = NMedals), position = position_stack(vjust = 0.5), size = 3, color = "white") +
  coord_flip() +
  labs(title = "Number of Medals by Sport (2016)",
       x = "Sport",
       y = "Number of Medals") +
  scale_fill_manual(values = c("Gold" = "gold", "Silver" = "grey", "Bronze" = "brown")) +
  theme_minimal()
```

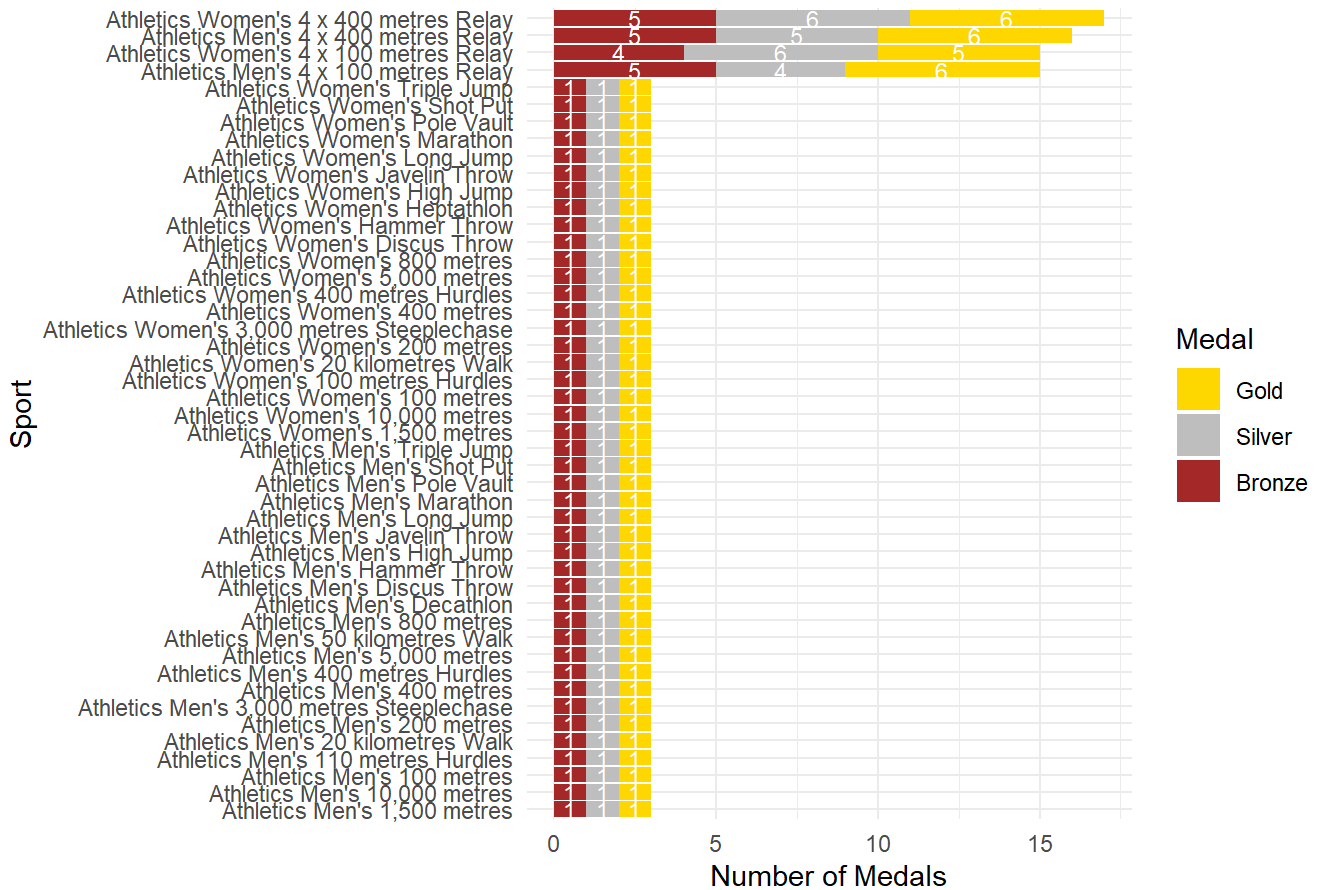
Number of Medals by Sport (2016)



are there more gold medals than silver or bronze in some sports e.g. athletics?

```
clean_data %>%
  filter(Sport == 'Athletics' & Year == 2016 & !(is.na(Medal))) %>%
  group_by(Event, Medal) %>%
  summarize(NMedals = sum(!is.na(Medal)), .groups = 'drop') %>%
  mutate(Medal = factor(Medal, levels = c("Gold", "Silver", "Bronze"))) %>%
  ggplot(aes(x = reorder(Event, NMedals), y = NMedals, fill = Medal)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = NMedals), position = position_stack(vjust = 0.5), size = 3, color = "white") +
  coord_flip() +
  labs(title = "Number of Medals by Sport (2016)",
        x = "Sport",
        y = "Number of Medals") +
  scale_fill_manual(values = c("Gold" = "gold", "Silver" = "grey", "Bronze" = "brown")) +
  theme_minimal()
```

Number of Medals by Sport (2016)



```
clean_data %>%
  filter(Event == "Athletics Women's 4 x 400 metres Relay" & Year == 2016 & Medal == 'Gold')
```

```
## AdjustedYear ID Sex Team Age Weight Height NOC Season Sport
## 1 2016 34551 F United States 30 56 168 USA Summer Athletics
## 2 2016 32382 F United States 23 65 183 USA Summer Athletics
## 3 2016 36677 F United States 24 72 180 USA Summer Athletics
## 4 2016 77390 F United States 27 68 171 USA Summer Athletics
## 5 2016 88474 F United States 22 59 173 USA Summer Athletics
## 6 2016 46542 F United States 30 68 173 USA Summer Athletics
## Event Medal AthleteCountry
## 1 Athletics Women's 4 x 400 metres Relay Gold USA
## 2 Athletics Women's 4 x 400 metres Relay Gold USA
## 3 Athletics Women's 4 x 400 metres Relay Gold USA
## 4 Athletics Women's 4 x 400 metres Relay Gold USA
## 5 Athletics Women's 4 x 400 metres Relay Gold USA
## 6 Athletics Women's 4 x 400 metres Relay Gold USA
## WeightHeightRatio Country Year HomeAway
## 1 0.3333333 Brazil 2016 Away
## 2 0.3551913 Brazil 2016 Away
## 3 0.4000000 Brazil 2016 Away
## 4 0.3976608 Brazil 2016 Away
## 5 0.3410405 Brazil 2016 Away
## 6 0.3930636 Brazil 2016 Away
```


Olympics Host Advange

Is there a significant advantage for countries playing at home vs away?

```
# Get the list of countries that have hosted
host_countries_lst <- unique(clean_data$Country)

# Loop through the list of host countries
for (country in host_countries_lst) {

  # Filter data for the current country
  medals_per_country <- clean_data %>%
    group_by(AdjustedYear, AthleteCountry) %>%
    summarise(Medals = sum(!is.na(Medal)), .groups = 'drop')

  hosting_years <- clean_data %>%
    filter(HomeAway == 'Home' & Country == country & Season == 'Summer') %>%
    pull(AdjustedYear) %>%
    unique()

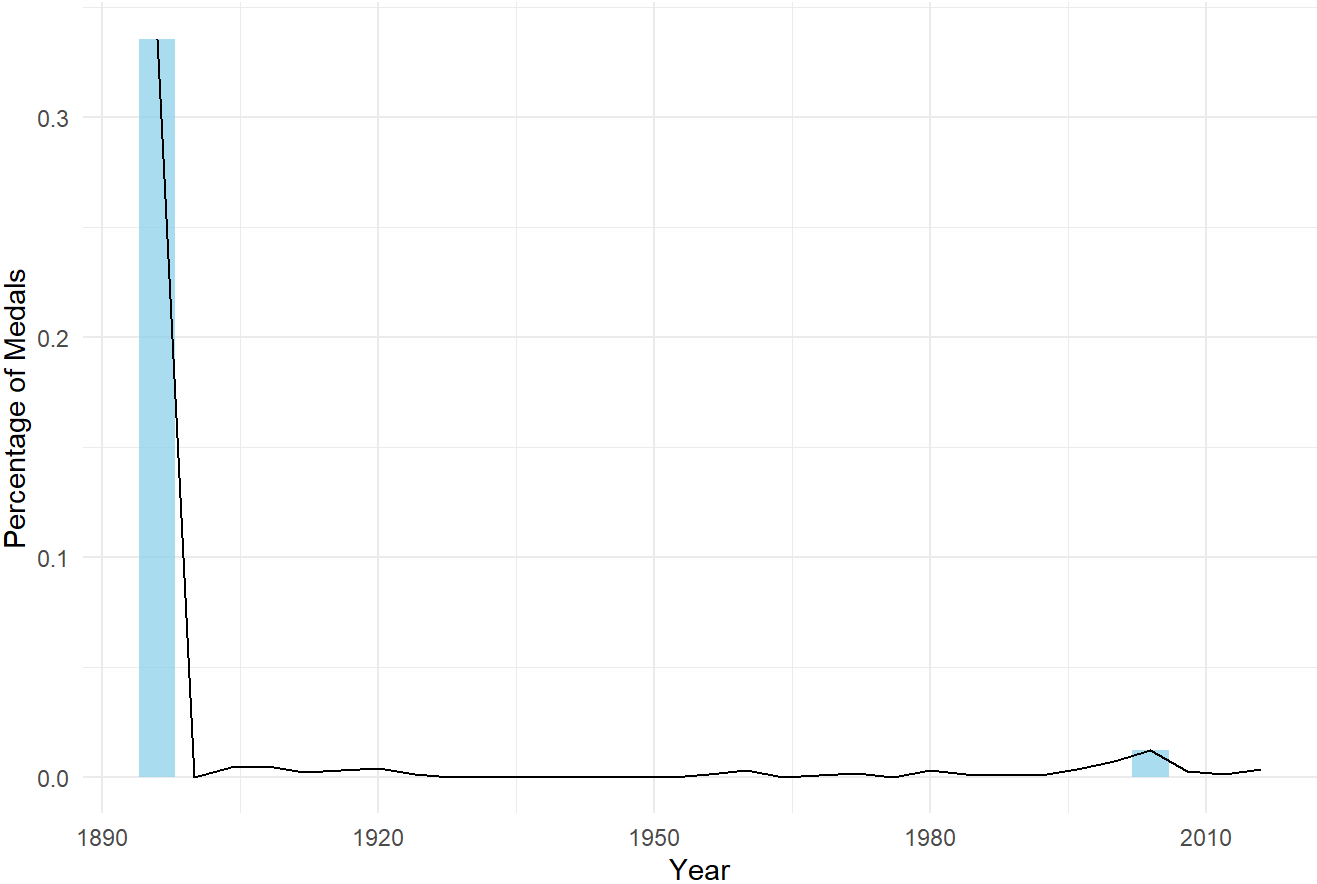
  plot_data <- clean_data %>%
    group_by(AdjustedYear) %>%
    summarize(TotalMedals = sum(!is.na(Medal)), .groups = 'drop') %>%
    left_join(medals_per_country, by = 'AdjustedYear') %>%
    mutate(MedalsPercent = Medals / TotalMedals) %>%
    filter(AthleteCountry == country)

  country_medals <- plot_data %>% filter(AdjustedYear %in% hosting_years)

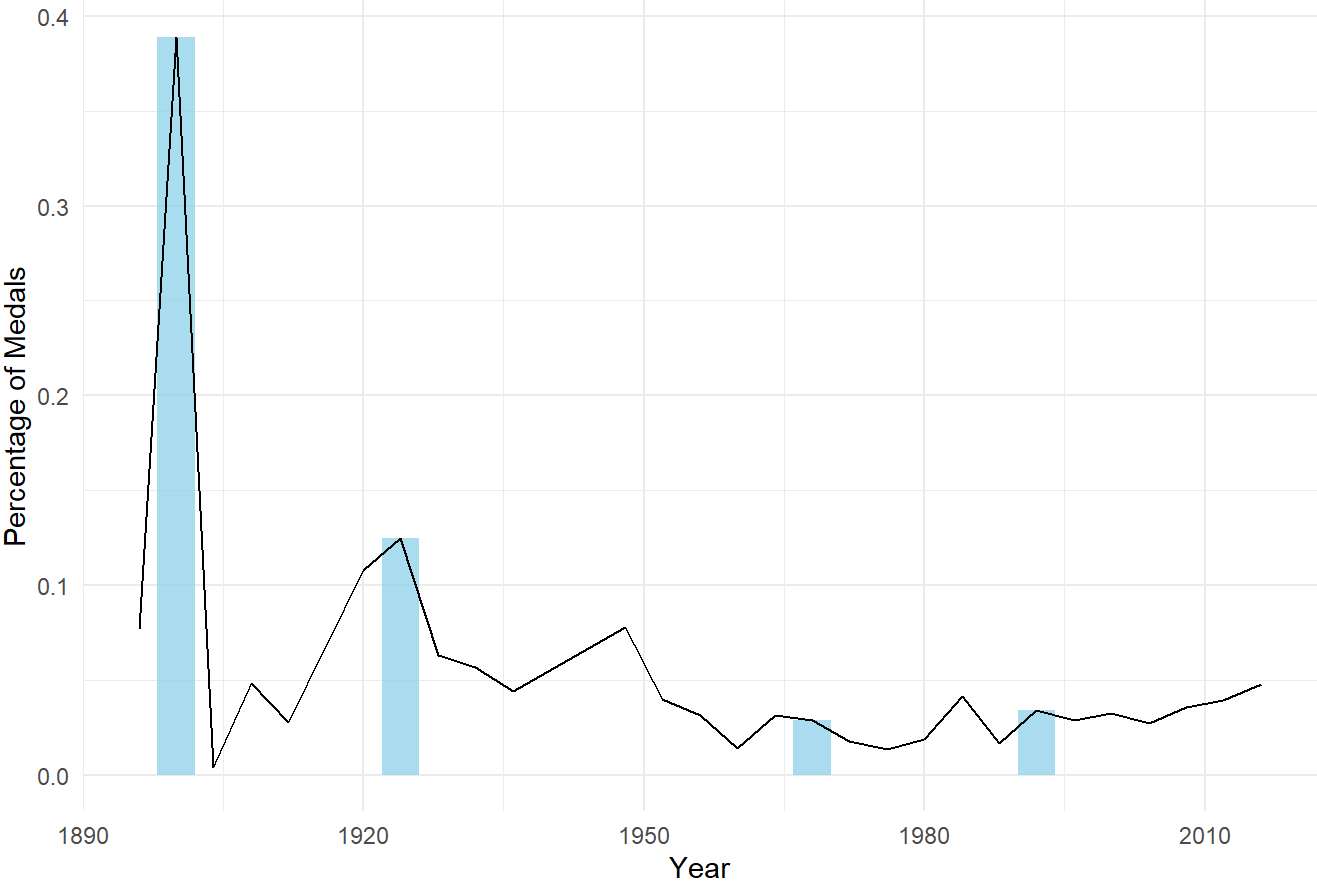
  # Create the plot
  p <- ggplot(data = plot_data, aes(x = AdjustedYear)) +
    geom_bar(data = country_medals, aes(y = MedalsPercent), stat = "identity", width = 4, fill =
"skyblue", alpha = 0.7) +
    geom_line(aes(y = MedalsPercent)) +
    labs(title = paste("Percentage of Medals Won Over Time -", country),
         x = "Year",
         y = "Percentage of Medals") +
    theme_minimal()

  # Print the plot for the current country
  print(p)
}
```

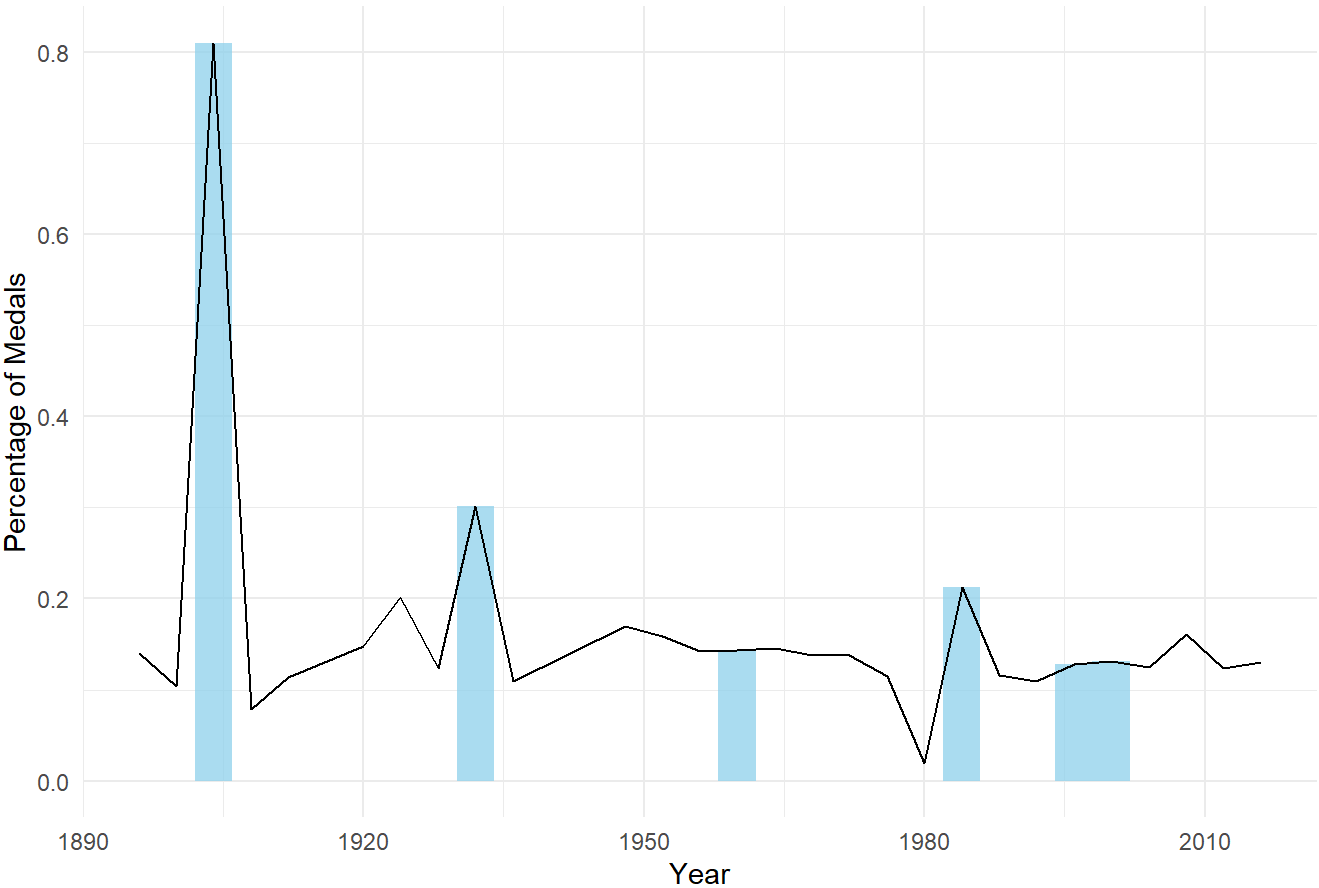
Percentage of Medals Won Over Time - Greece



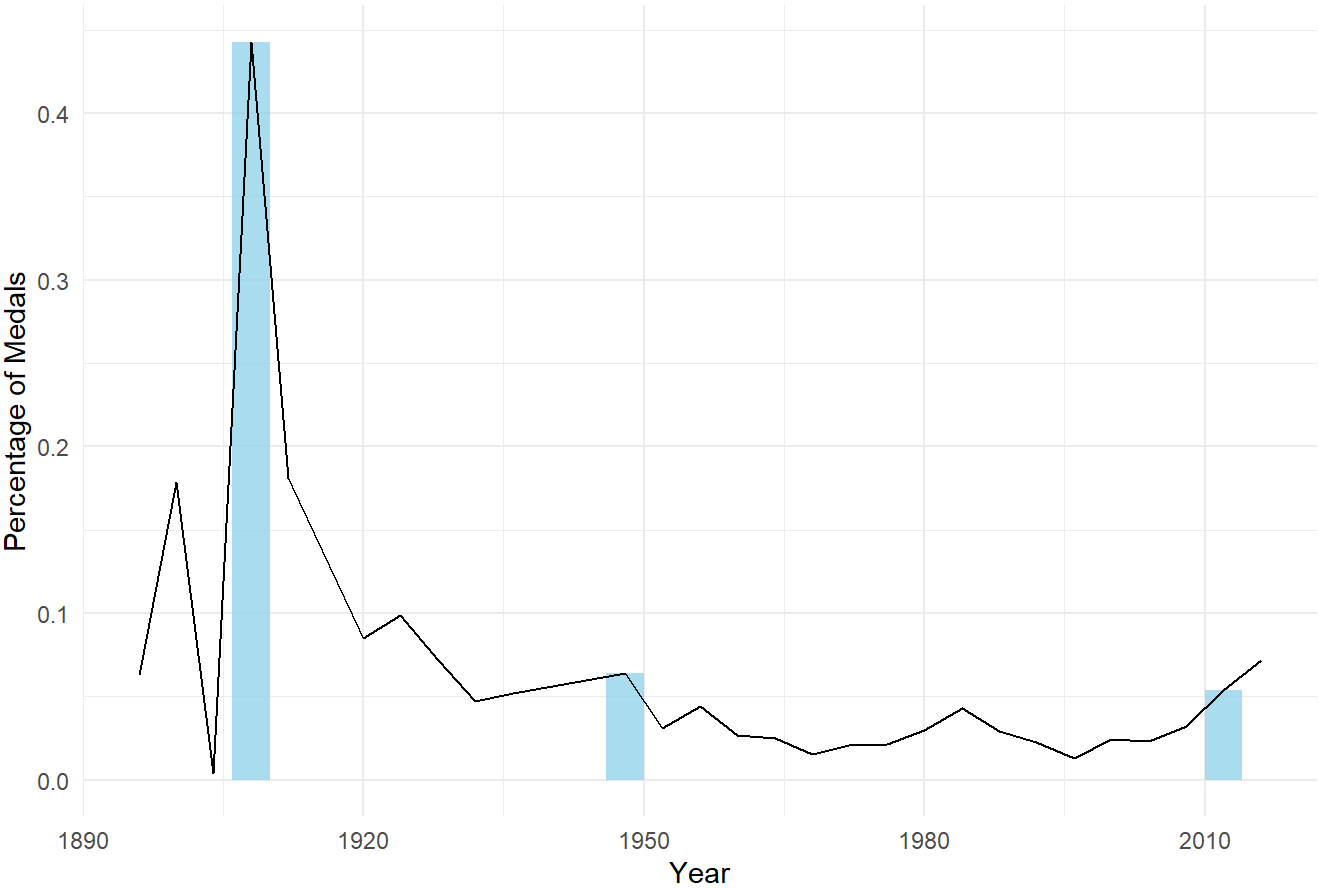
Percentage of Medals Won Over Time - France



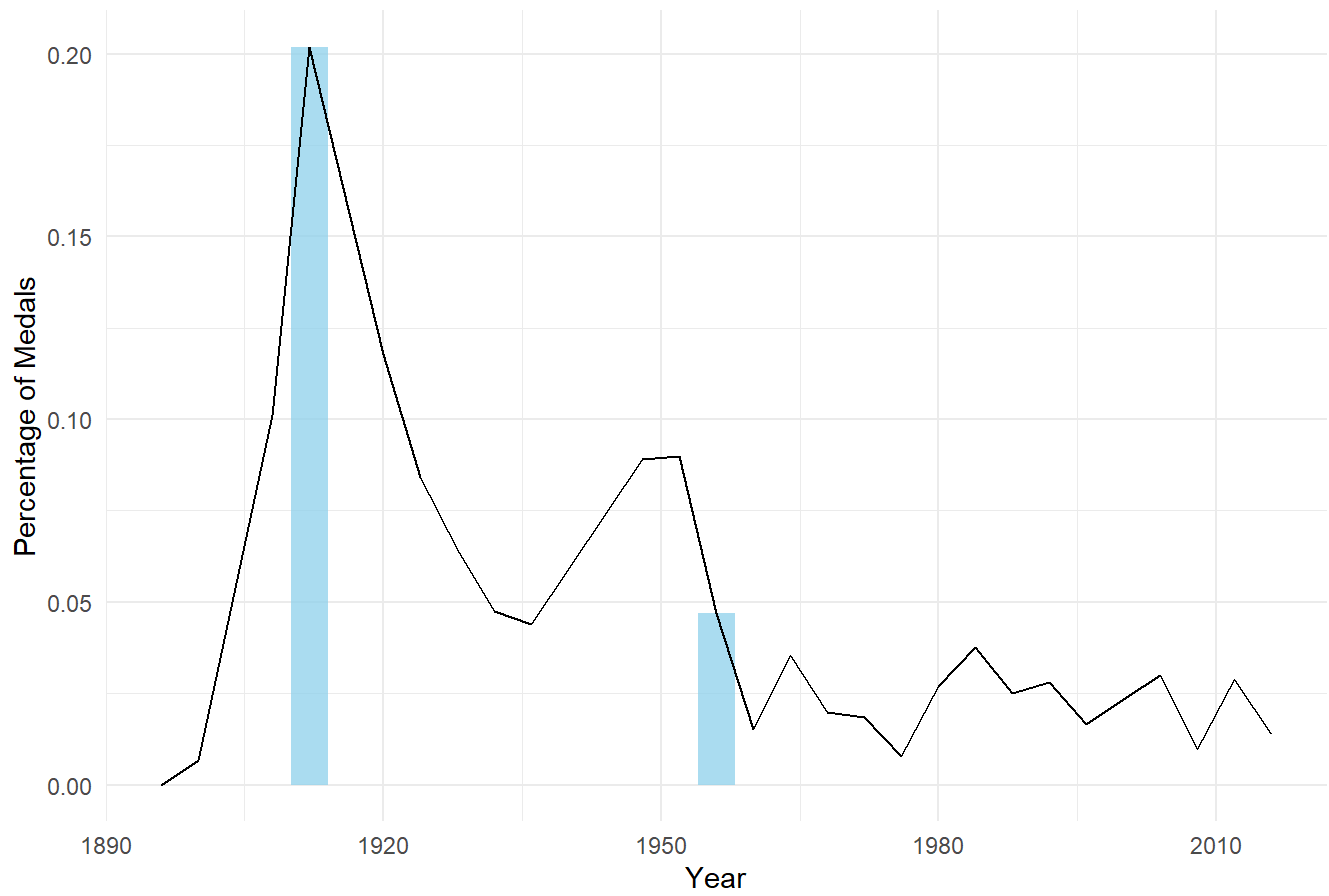
Percentage of Medals Won Over Time - USA



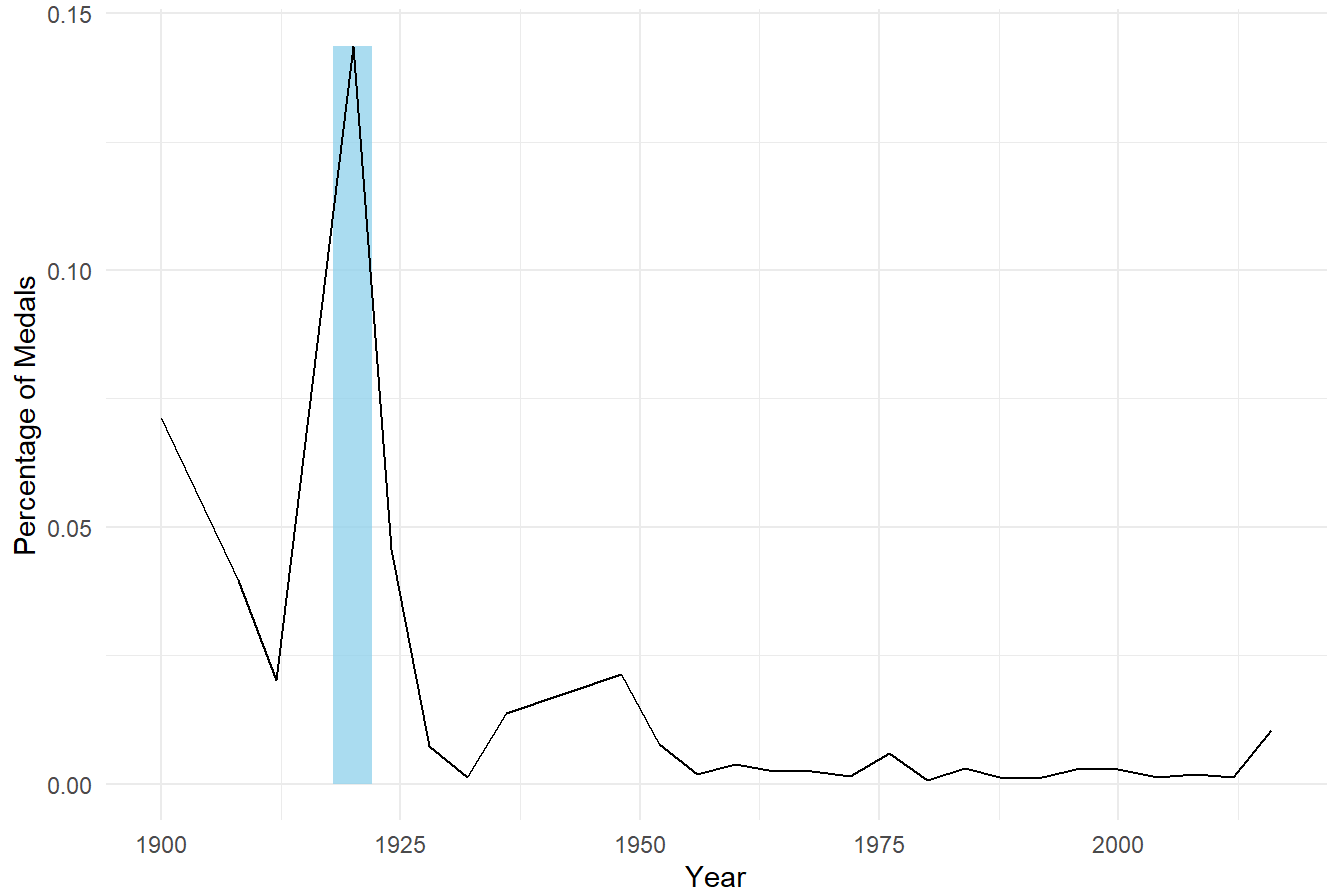
Percentage of Medals Won Over Time - UK



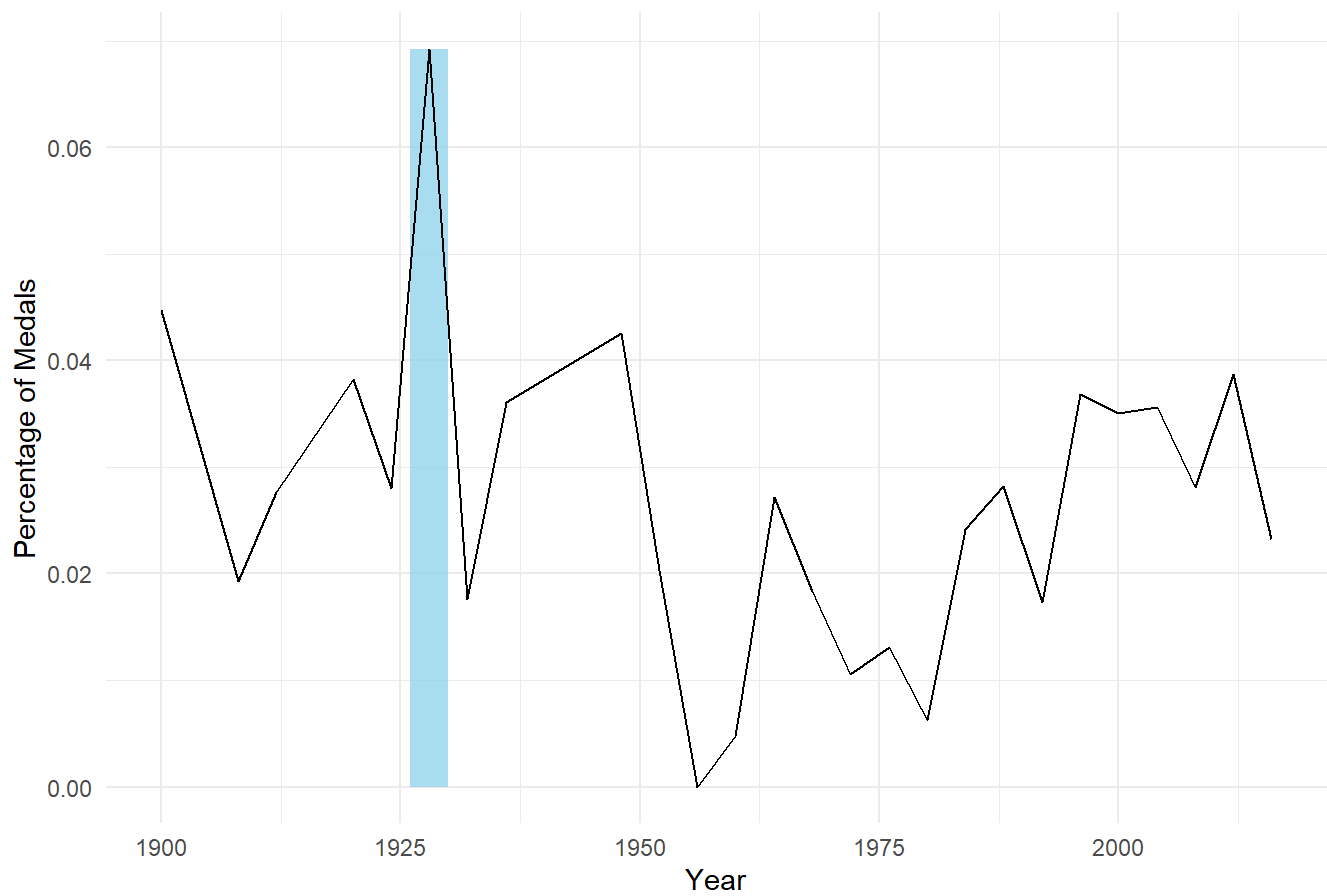
Percentage of Medals Won Over Time - Sweden



Percentage of Medals Won Over Time - Belgium



Percentage of Medals Won Over Time - Netherlands



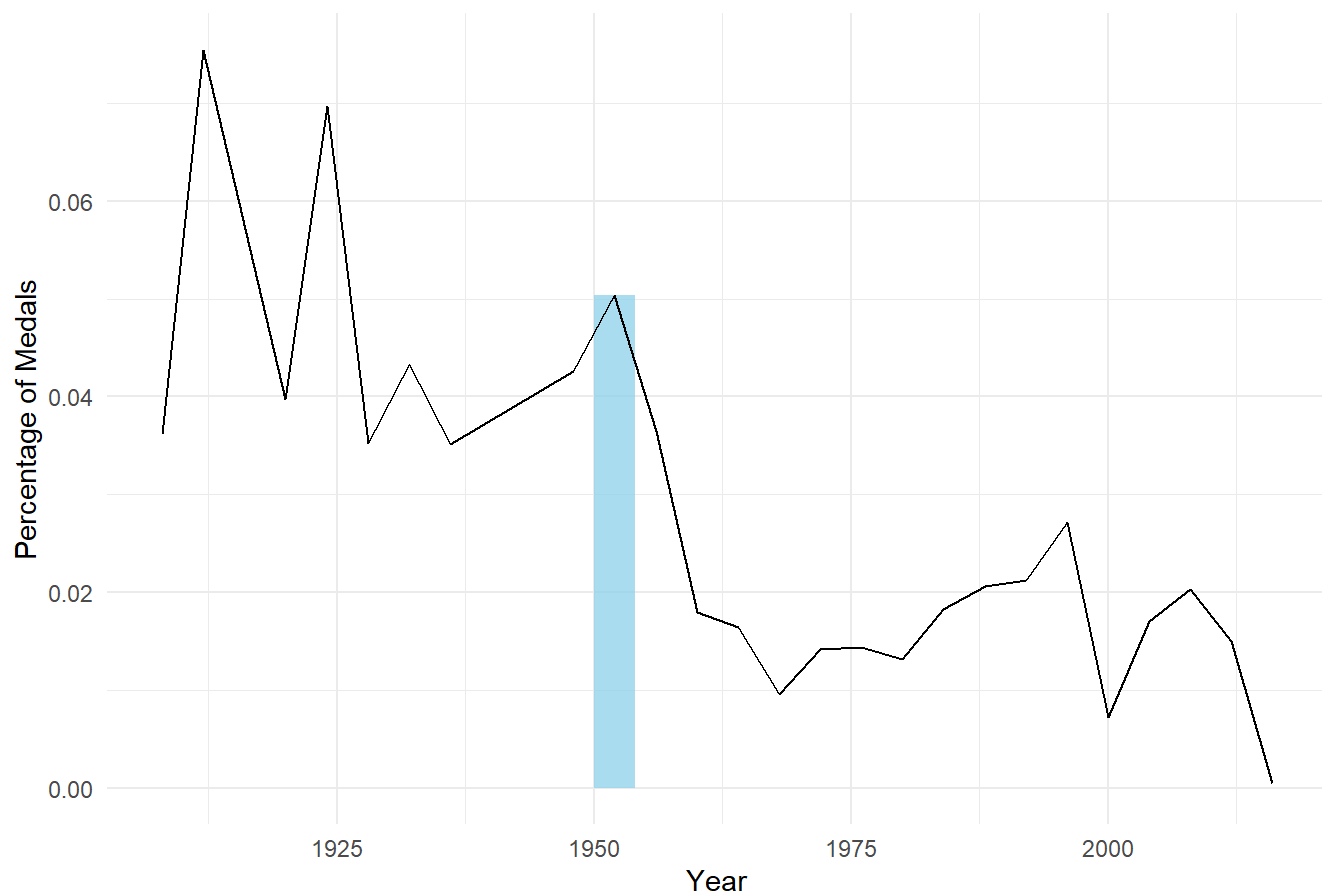
Percentage of Medals Won Over Time - Switzerland



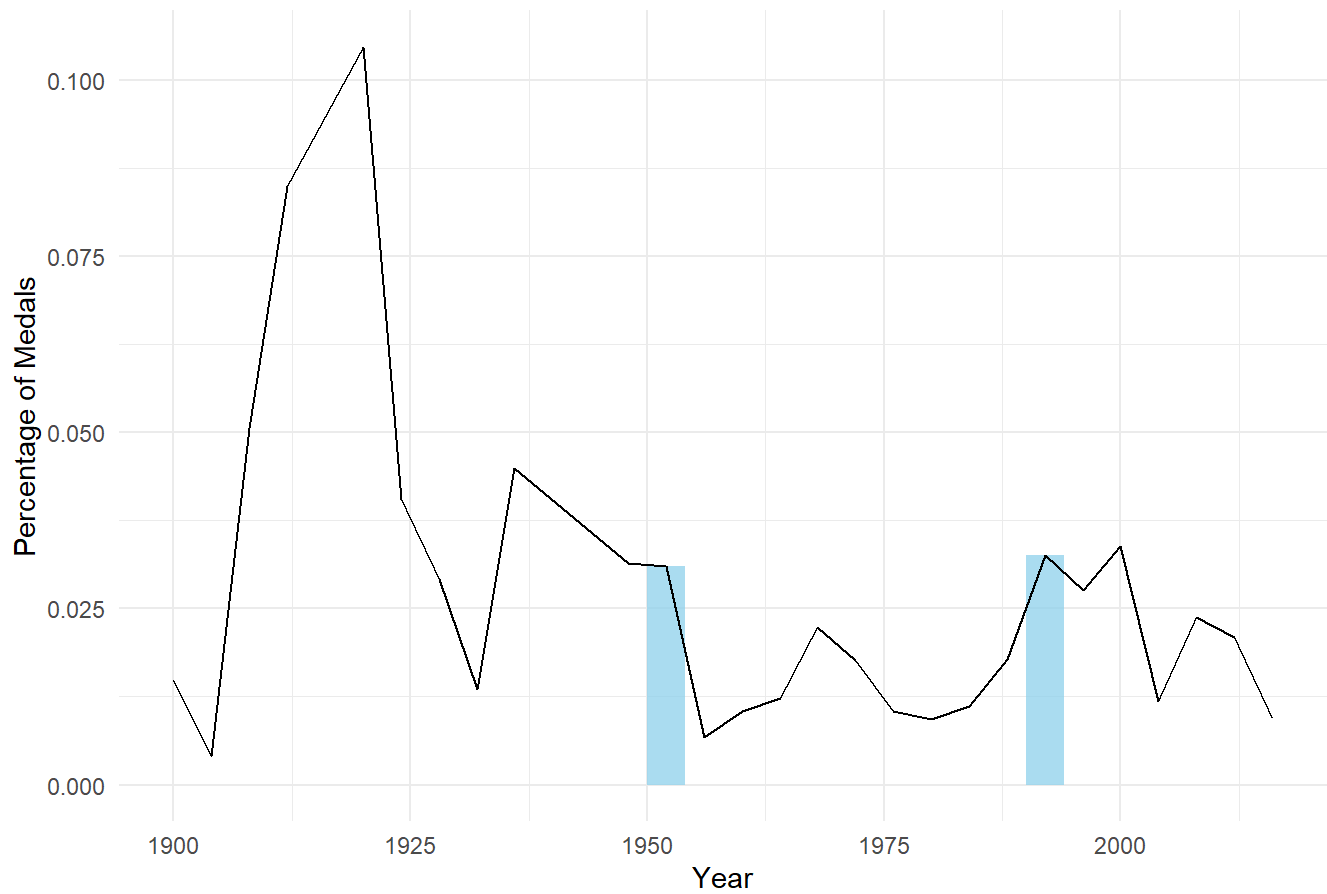
Percentage of Medals Won Over Time - Germany



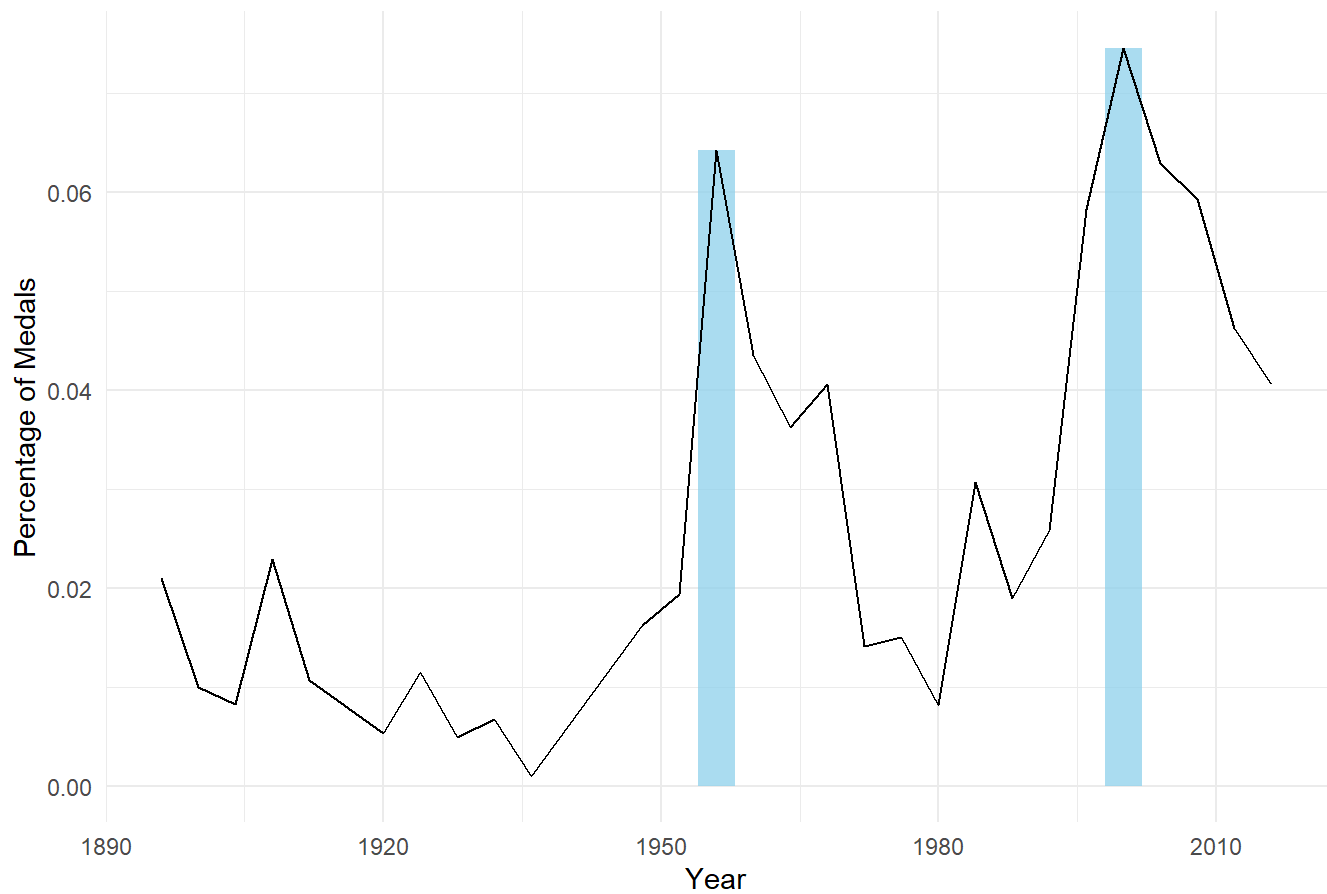
Percentage of Medals Won Over Time - Finland



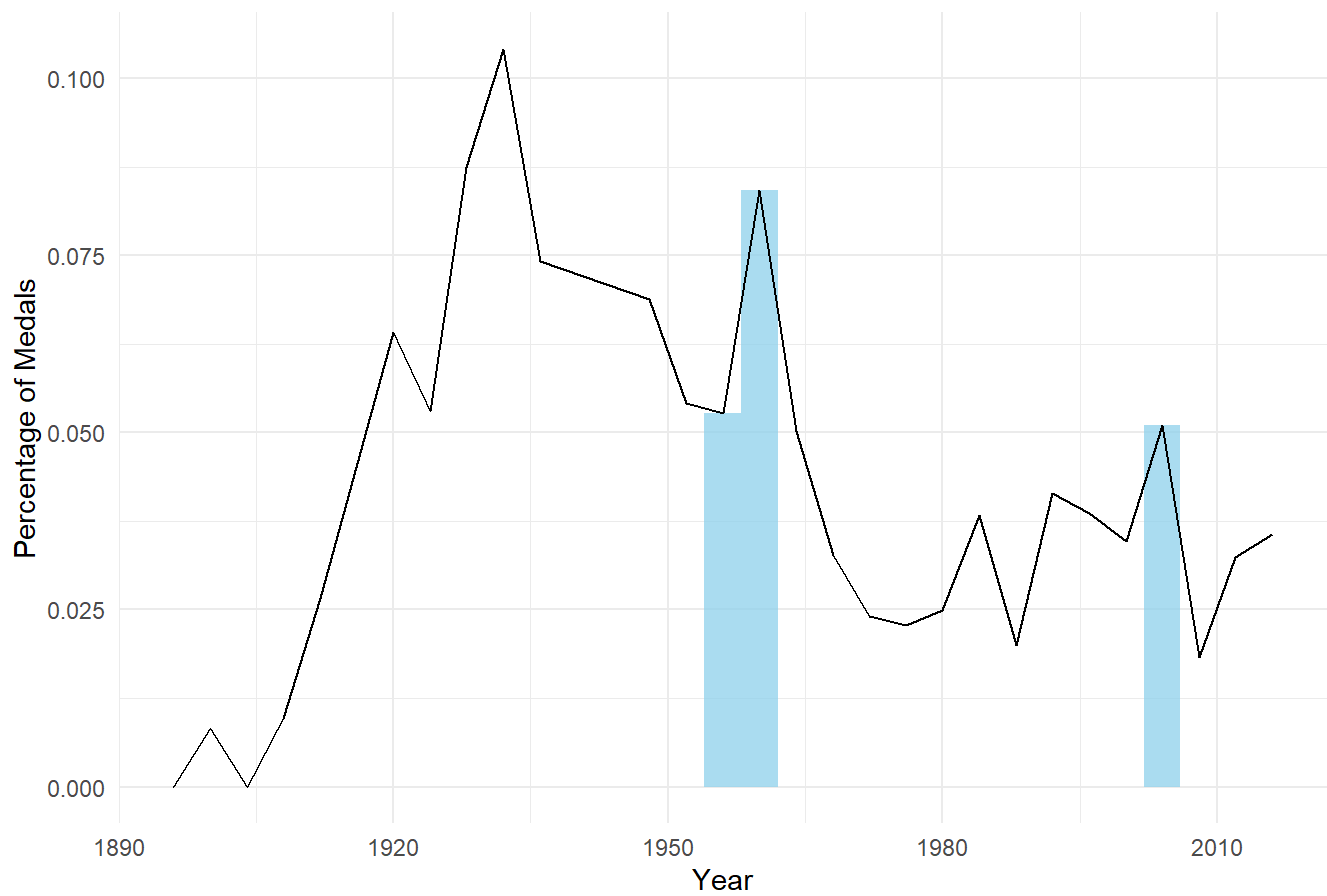
Percentage of Medals Won Over Time - Norway



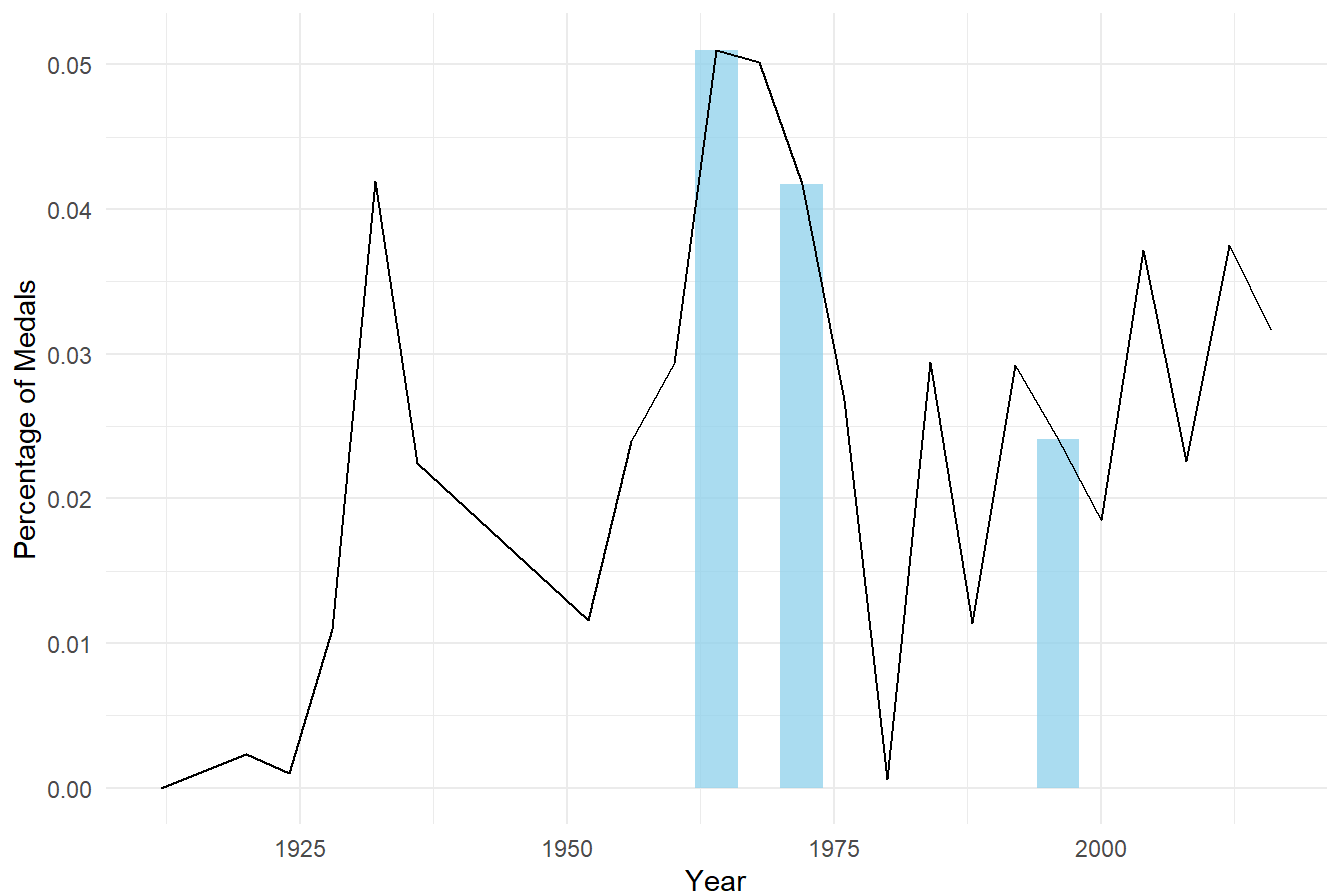
Percentage of Medals Won Over Time - Australia



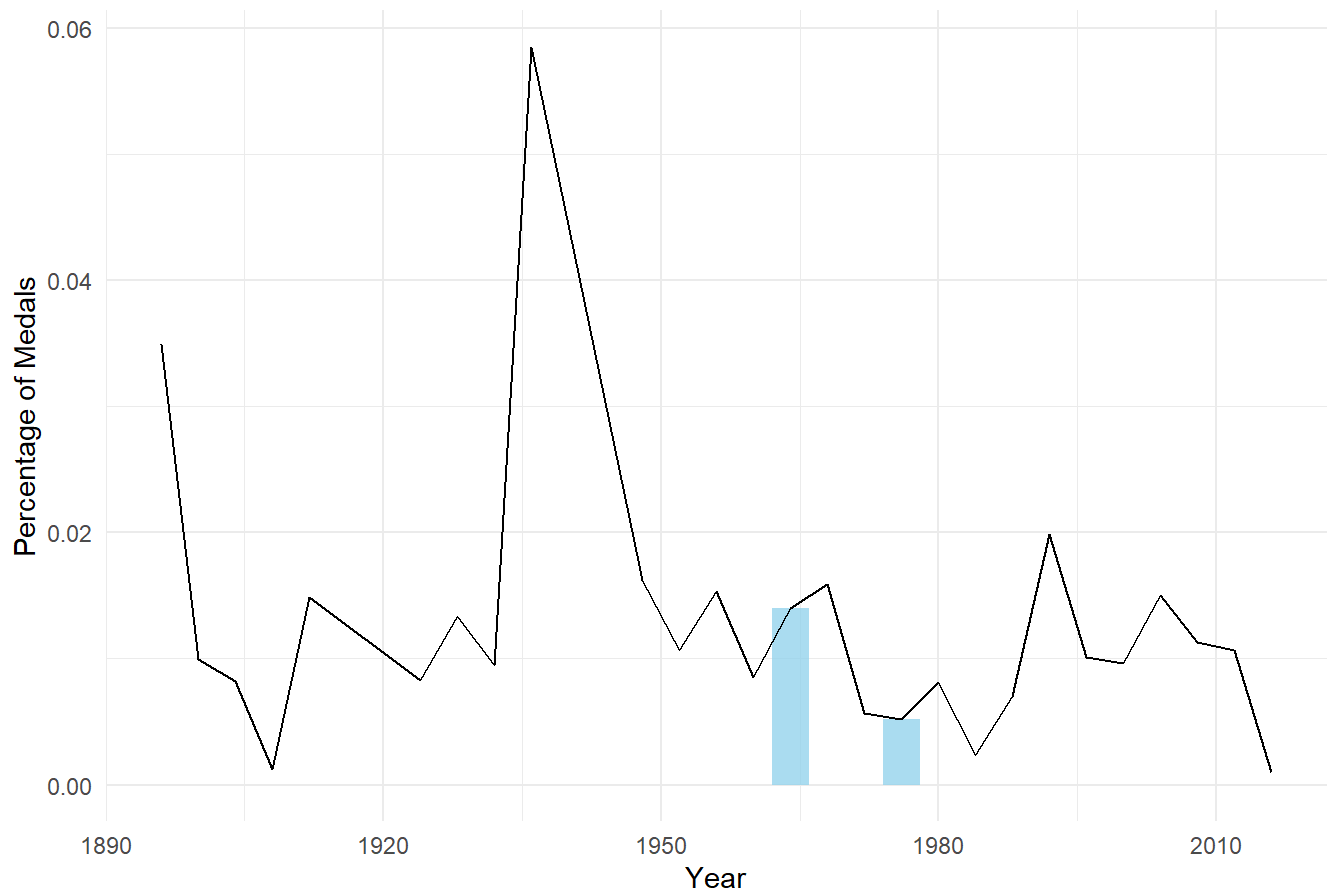
Percentage of Medals Won Over Time - Italy



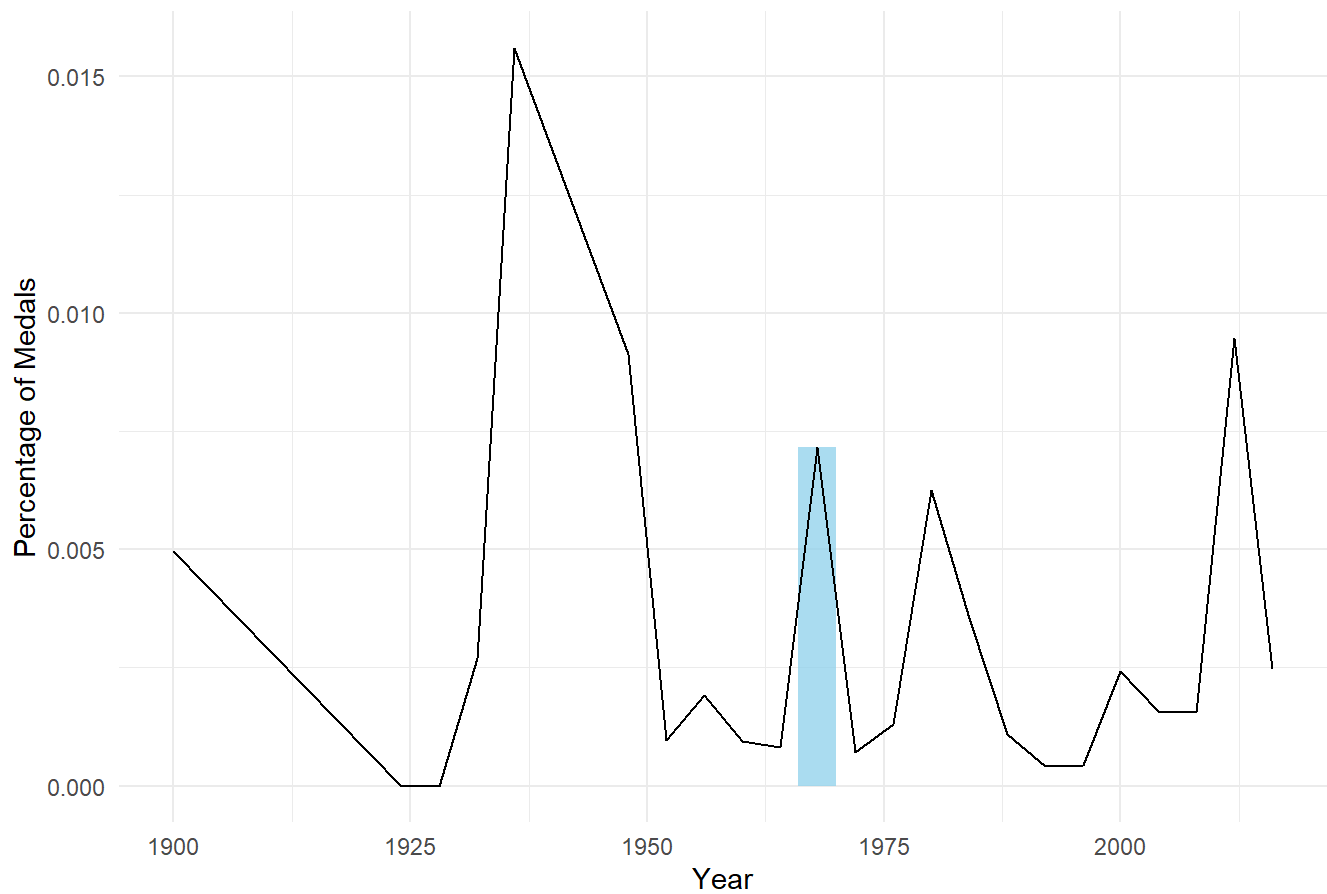
Percentage of Medals Won Over Time - Japan



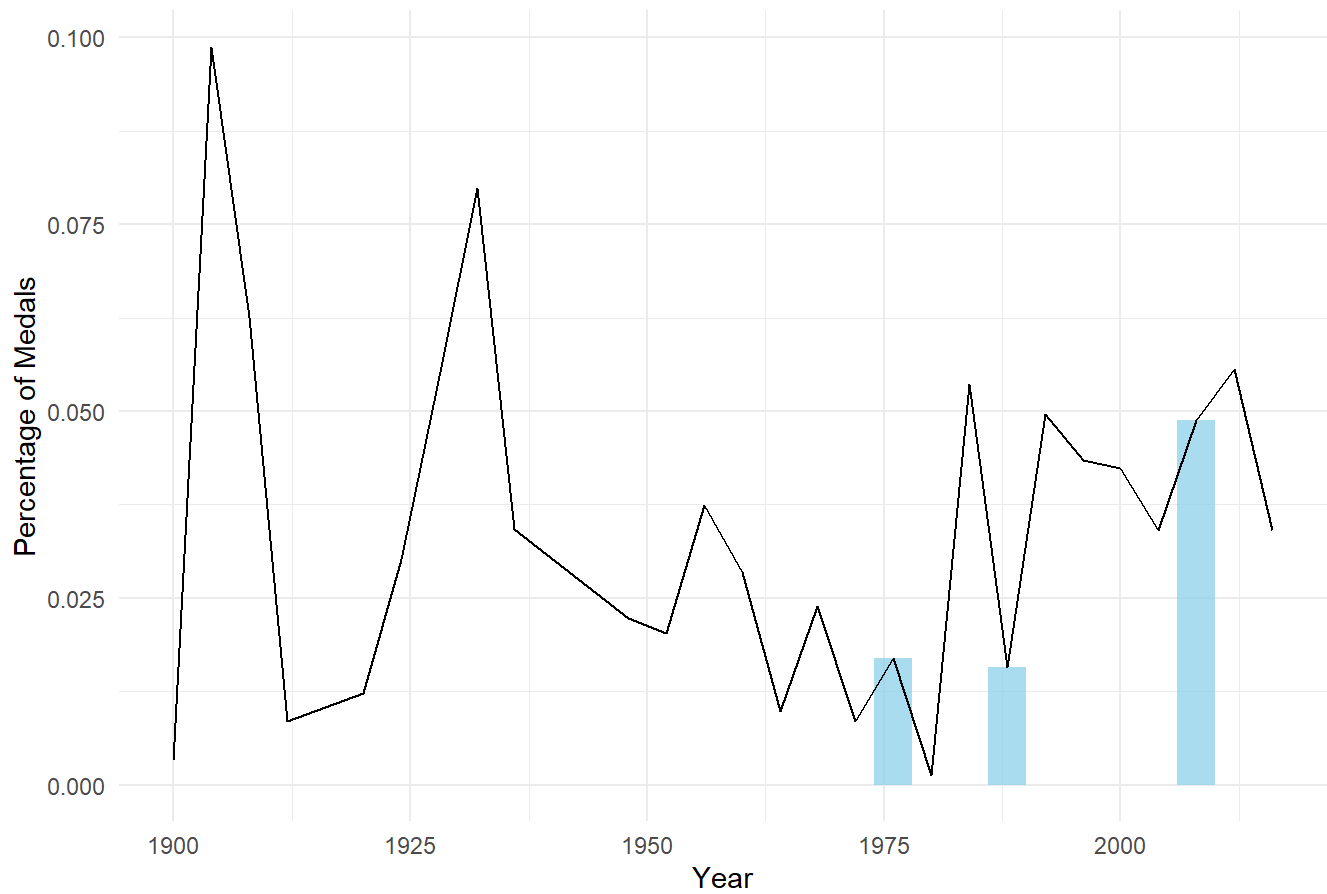
Percentage of Medals Won Over Time - Austria



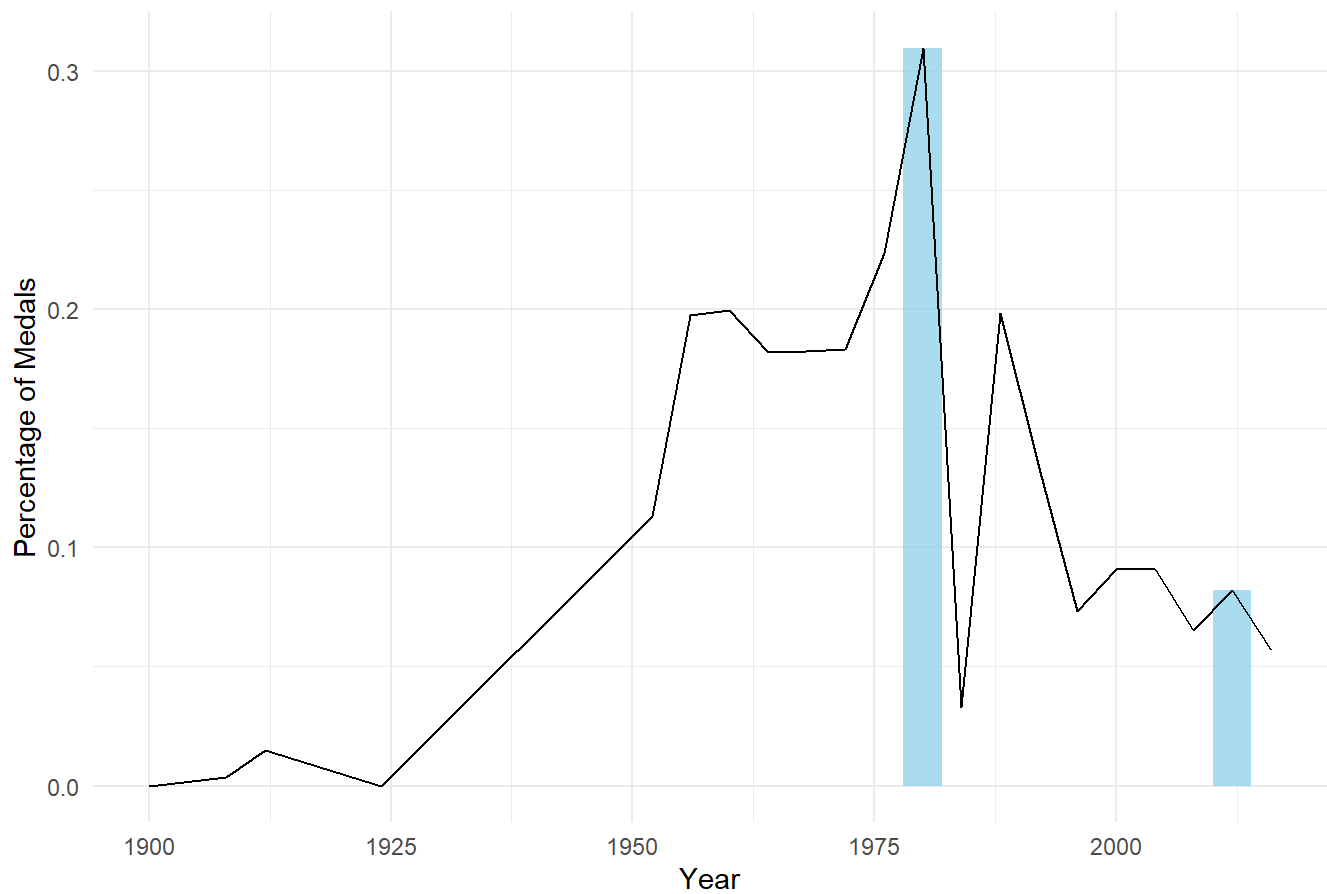
Percentage of Medals Won Over Time - Mexico



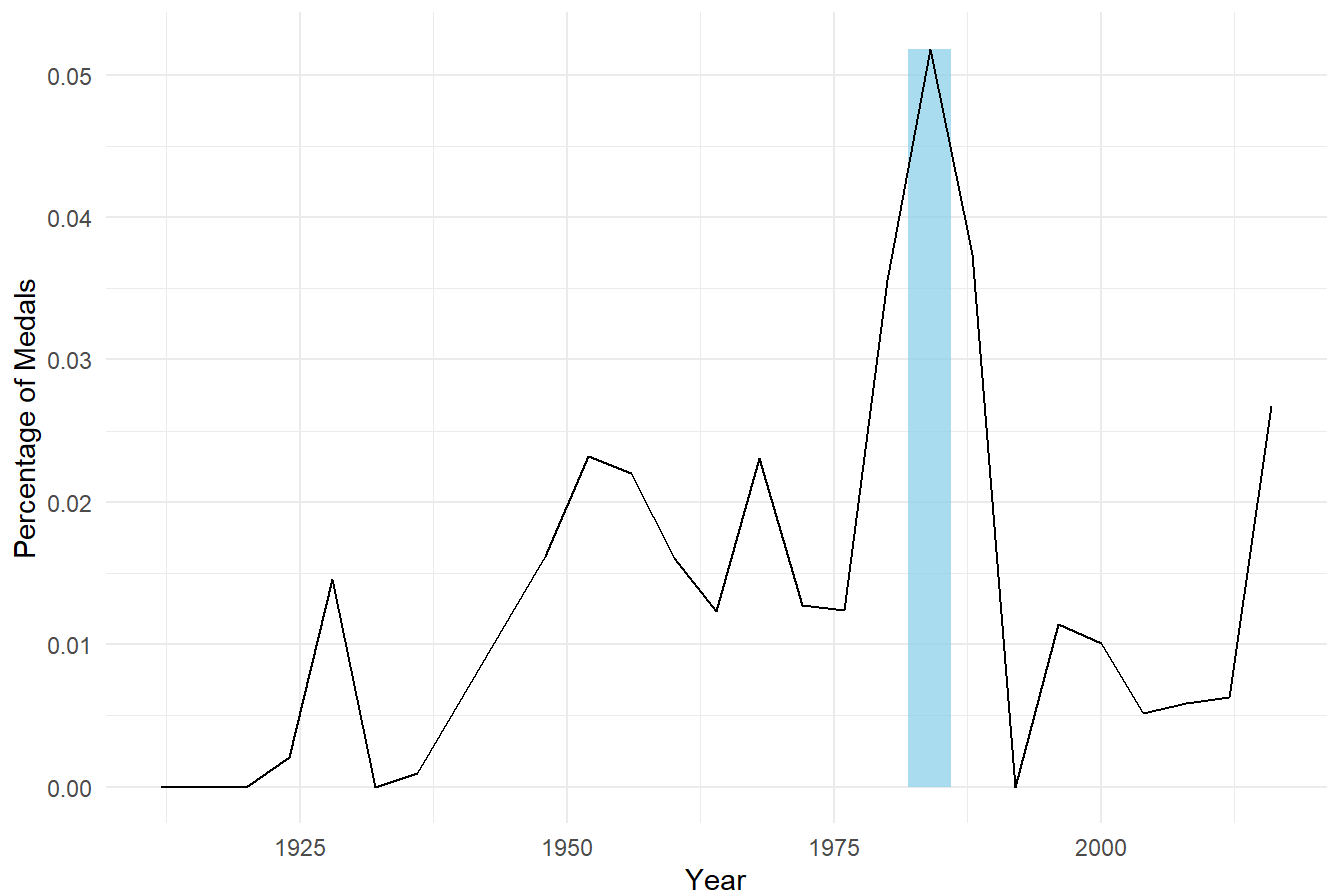
Percentage of Medals Won Over Time - Canada



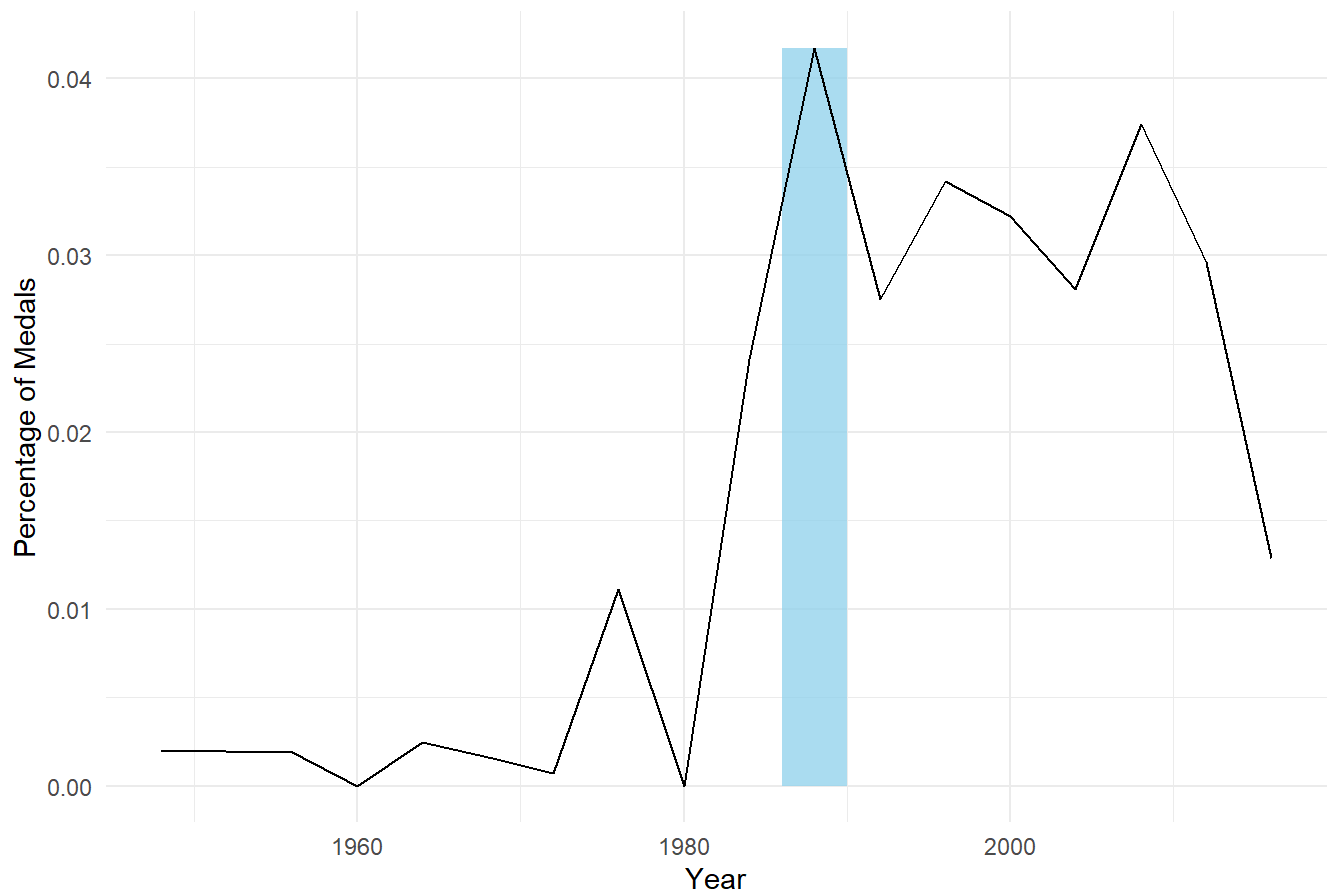
Percentage of Medals Won Over Time - Russia



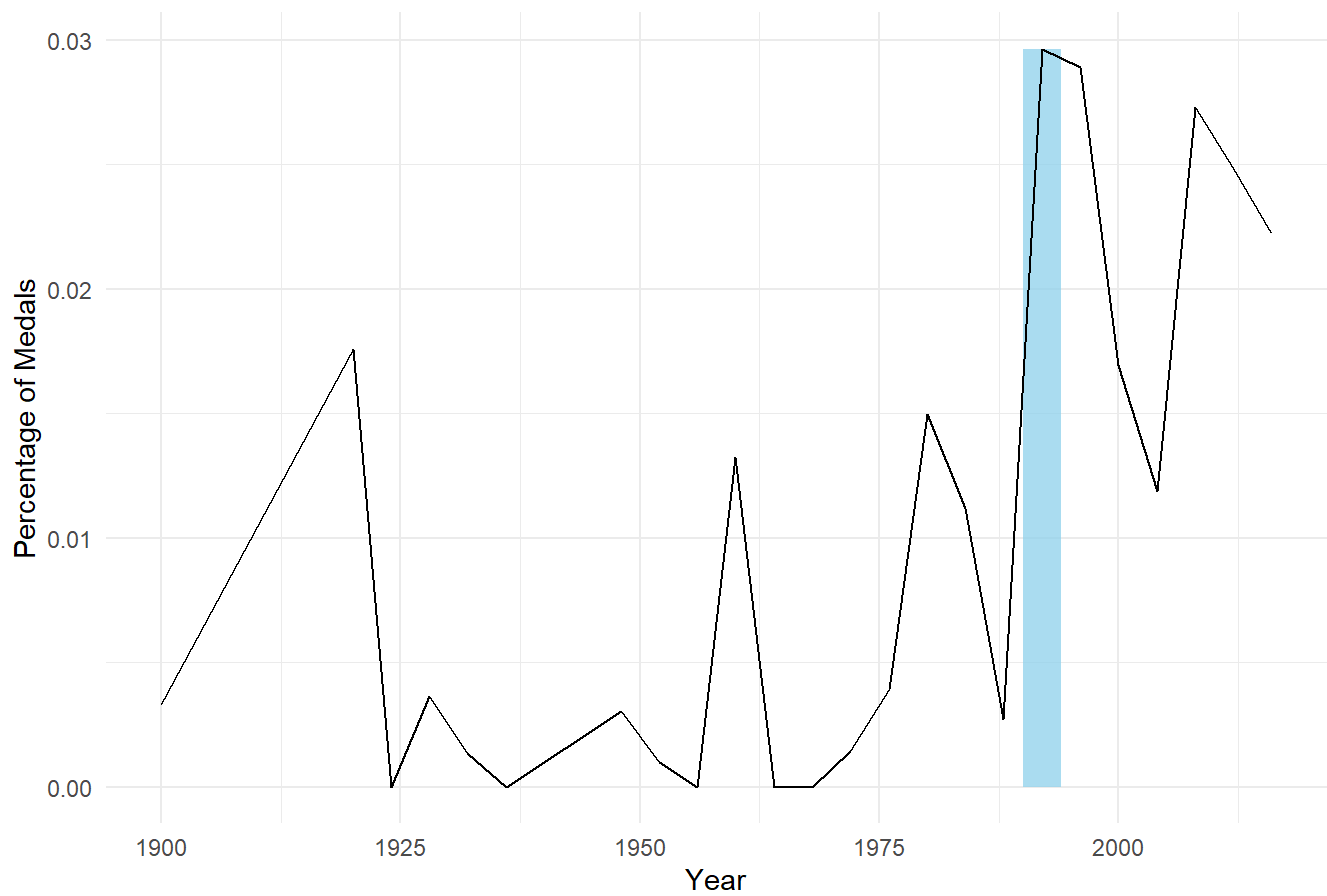
Percentage of Medals Won Over Time - Serbia



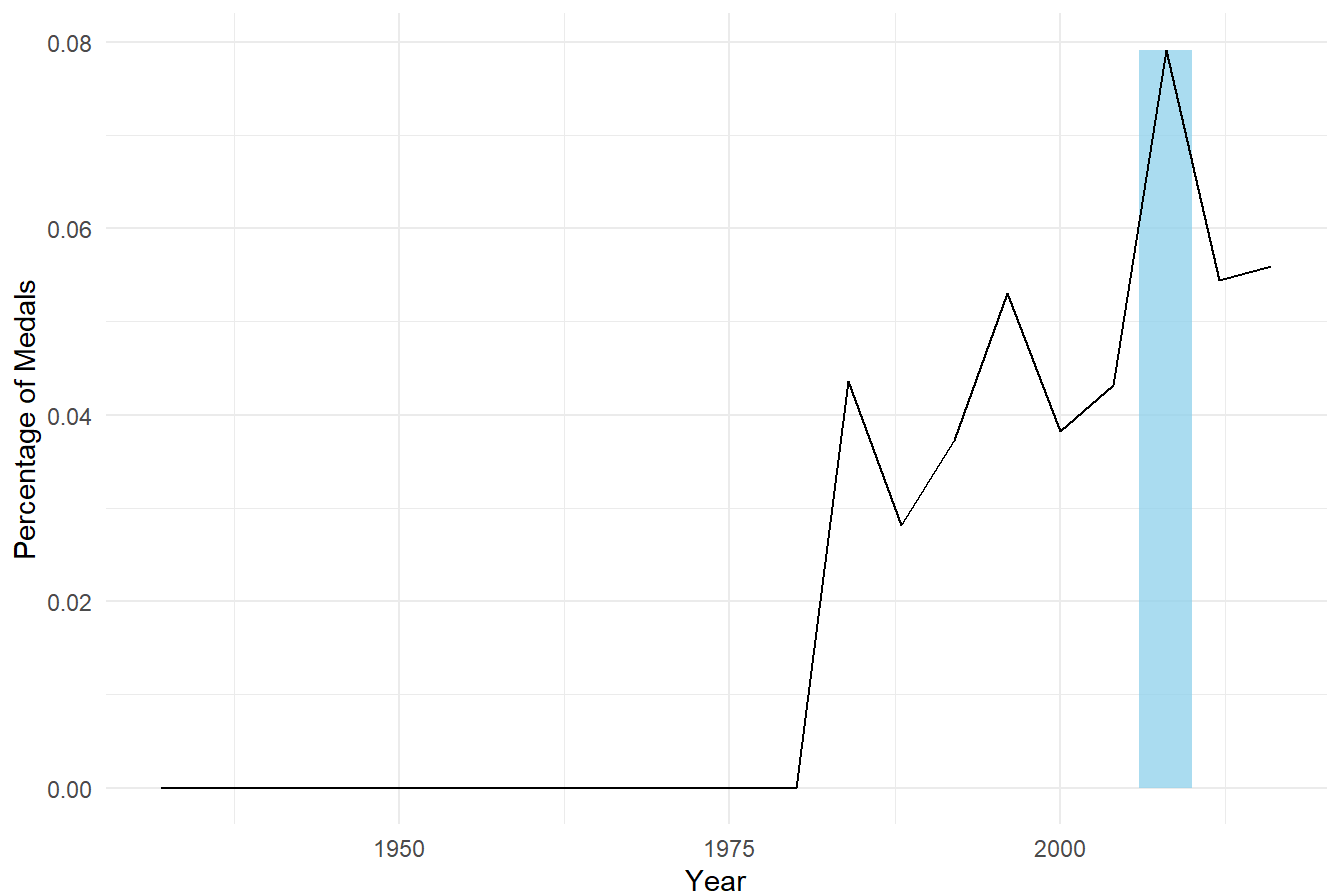
Percentage of Medals Won Over Time - South Korea



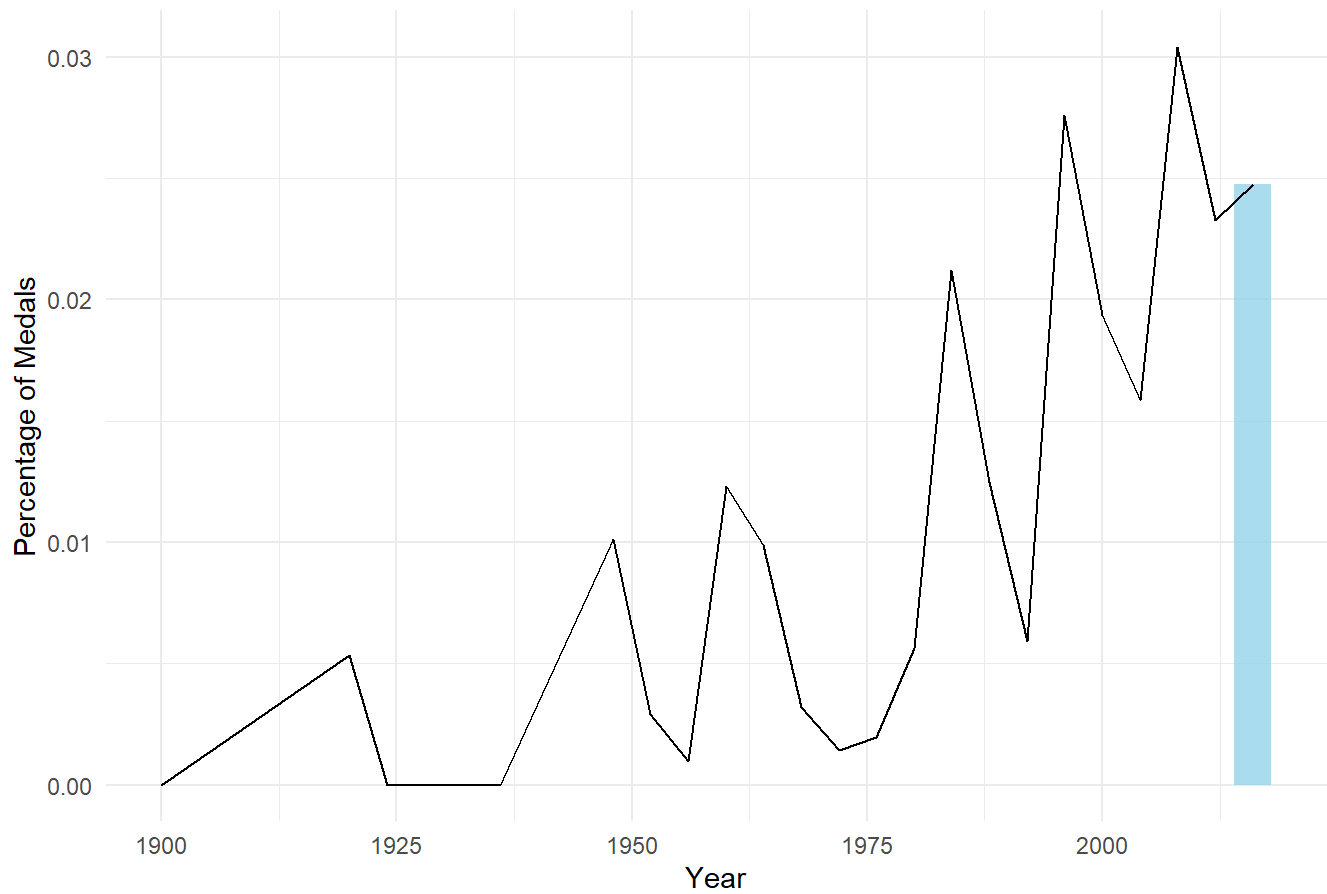
Percentage of Medals Won Over Time - Spain



Percentage of Medals Won Over Time - China



Percentage of Medals Won Over Time - Brazil



Correlation of medals to GDP or Population